Technical Validation Supplement

Measurement of $\nu_\mu + \bar{\nu}_\mu$ Charged-Current Single Pion Production on Argon in the NuMI Beam

Companion to the analysis note

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Abstract

This supplement holds the validation and diagnostic material for the CC$\pi^\pm$ NuMI cross-section measurement: the studies that establish what the measurement can and cannot resolve, the estimators that were tried and rejected, and the closure tests that qualify the extraction. It is separated from the analysis note so that the note describes one current analysis state, while the record of how that state was arrived at remains available and citable.

Nothing here is superseded by the note; the two are consistent by construction, having been split from a single source. Material that *was* superseded — the corrections that changed a result — is recorded separately in the change log, with before-and-after numbers.

The central results of this supplement are negative ones, and they set the scope of the measurement. Charged pions interact hadronically before ranging out, so the reconstructed momentum saturates near 0.34 GeV/c and carries little information about the initial momentum above roughly 0.4 GeV/c. Every alternative estimator tried — golden-pion tagging, multiple Coulomb scattering, calorimetry, multivariate regression, daughter-energy recovery, and the proton-tagged selection — fails to recover it, and a truth-level golden selection performs no better than the classifier, which places the limitation in the physics rather than in the estimator. This is what reduces p_π to two bins and $W_{\pi p}$ to an integrated cross section.

Contents

1 Pion momentum: the range estimator and why it fails

1.1 The pion-momentum bias and why p_π is the worst-conditioned observable

The p_π response is the least diagonal of any observable in the analysis. This section quantifies why, because the cause is physical and sets a floor on what the measurement can say about the pion momentum spectrum.

Two separate effects, only one of them corrected. The reconstructed pion momentum stored in the ntuple, `candidate_pion_mom_reco`, is `trk_range_muon_mom_v`: a range-based momentum evaluated under the *muon* mass hypothesis and applied to a pion track. Two distinct things follow.

First, the mass hypothesis itself biases the estimate low, because for a given range a pion carries more momentum than a muon. The size is ∼17–19% from the CSDA range scaling, and it is essentially independent of momentum. This effect *is* corrected, though not in the branch: the reco bin definitions in `ccpi_ppi_bin_config_opt.txt` apply a kinetic-energy-preserving $\mu \to \pi$ mass swap, $E_\mu \to T \to E_\pi \to p_\pi$, before binning. It works; in the lowest true bin $\langle p^{\rm reco}/p^{\rm true}\rangle$ goes from 0.888 with the raw branch to 0.976 with the correction.

Second, and not corrected by anything, charged pions interact hadronically before coming to rest. The reconstructed track then ends at the interaction point rather than the stopping point, so the visible range, and any momentum inferred from it, is truncated. This grows rapidly with momentum and comes to dominate.

The reconstructed momentum saturates. Table 1 and Figure 1 make the consequence plain: as the true momentum rises by a factor of four and a half, the reconstructed momentum barely moves, saturating at ≈ 0.34 GeV/c. Above that the estimator carries almost no information about how energetic the pion actually was.

Table 1: Pion momentum reconstruction versus truth, selected signal, Run-1 FHC. "raw" is the `candidate_pion_mom_reco` branch; "corr." additionally applies the $\mu \to \pi$ mass swap used by the reco bin definitions. The mass correction removes the bias at threshold; nothing removes the saturation above it.

true p_π [GeV/c]	N	$\langle p^{\rm true}\rangle$	$\langle p^{\rm corr}\rangle$	raw/true	corr./true
0.175–0.250	528	0.210	0.205	0.888	0.976
0.250–0.300	270	0.276	0.238	0.790	0.863
0.300–0.350	254	0.325	0.253	0.716	0.779
0.350–0.400	265	0.374	0.270	0.664	0.721
0.400–0.500	403	0.443	0.279	0.581	0.629
0.500–0.600	252	0.548	0.307	0.520	0.560
0.600–0.800	272	0.688	0.326	0.441	0.473
0.800–1.200	164	0.951	0.336	0.329	0.353

What that does to the response matrix. Migrating the five analysis bins with the corrected estimator gives a nearly lower-triangular matrix: the diagonal fraction falls from 94.3% in the first true bin to 36.5, 23.1, 10.4 and finally 6.2% in the last. For the highest true bin ($p_\pi > 0.55$ GeV/c) only 6.2% of events are reconstructed in the corresponding reco bin while 32.6% fall all the way into the *lowest* one. Every true bin above the first therefore deposits most of its events into reco bin 1, which is exactly the "pile-up into the first bin" seen in Fig. 18 of the analysis note.

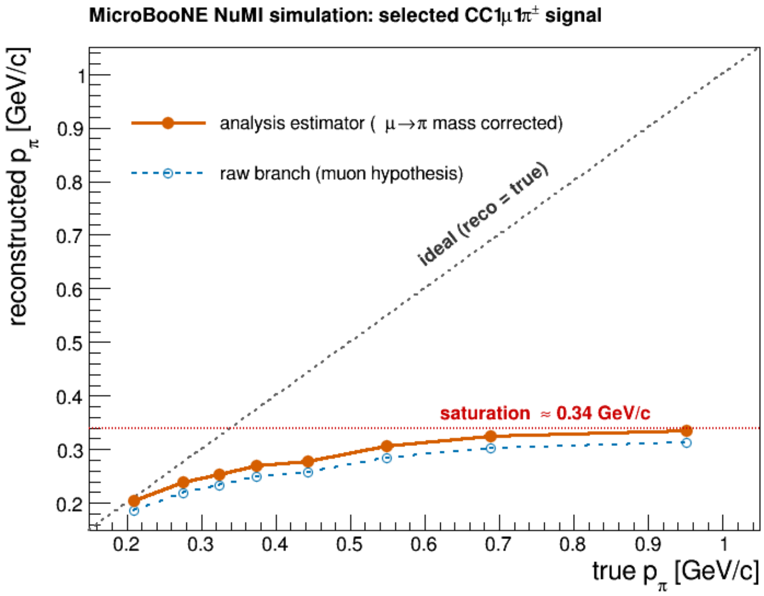


Figure 1: Mean reconstructed against mean true pion momentum, selected signal. The analysis estimator (filled, $\mu \to \pi$ mass corrected) tracks truth at threshold but flattens out near 0.34 GeV/c: a true 0.95 GeV/c pion reconstructs at the same place as a true 0.55 GeV/c one. The raw `candidate_pion_mom_reco` branch (open) sits systematically below it by the uncorrected mass-hypothesis offset. The saturation is caused by hadronic interaction truncating the track range, and no range-based estimator can recover it.

How the BNB CC1$\pi^\pm$ analysis handles the same problem. The published BNB CC1$\pi^\pm$Xp analysis (internal note v0.91) faces an identical estimator, it too takes "the momentum of pion candidates from range", and its treatment is worth reproducing here, because it is a solved problem there and an open one here.

That analysis does three things we do not. First, it defines a *golden pion*: one that comes to rest and leaves a Bragg peak, as opposed to one that scattered or interacted. Second, it trains a dedicated *golden-pion BDT* and applies it as an extra cut for the pion-momentum differential cross section only, leaving the total and angular results on the generic selection. The BDT separates golden from non-golden pions using track "wiggliness" (the standard deviation of the angular difference between directions of adjacent points along the fitted track, so a direct handle on scattering), the number of reconstructed descendant particles, the track–shower score, and the Bragg likelihood ratio. The cost and benefit are quoted explicitly: efficiency falls from 20.3% to 9.5% and purity is essentially unchanged ($51.9 \to 49.5\%$), but the golden fraction of the selected signal rises from 36.3% to 67.7%. They apply the same logic to the muon momentum, restricting that cross section to contained muons (8.4% efficiency, 56.1% purity) so it uses range only.

Third, and most directly comparable, their binning is chosen against an explicit criterion: each bin must hold at least 100 predicted MC events *and* the diagonal element of the smearing matrix must exceed 0.68. The resulting pion-momentum edges are $0.10, 0.16, 0.19, 0.22, 0.60$ GeV/c with an overflow bin, fine structure only where the estimator still tracks truth, then a single wide bin from 0.22 to 0.60 and everything above 0.60 in overflow.

Set against that criterion our binning does not survive: the diagonal fractions measured above are $94.3, 36.5, 23.1, 10.4$ and 6.2%, so *one* of five bins passes 0.68 and the top three fail it by a wide margin. Our edges (0.175–0.550 in five bins, open above) put four boundaries inside precisely the region where the reconstructed momentum has saturated.

Consequences, and what would improve it.

Consequences, and what would improve it. This is not a bias on the extracted cross section; the response matrix contains it and the unfolding inverts it, but it is the reason the p_π result leans harder on the regularisation than any other observable, and it explains the A_C survey result that the momentum observables retain only 11–21% of their truth-level model discrimination (Table 10) while $\theta_{\mu\pi}$ retains 58–93%. With 6% diagonal strength in the top bin, the measurement above $\sim$0.5 GeV/c is effectively inferred from the response model rather than measured.

The BNB note therefore supplies a tested recipe, and *all four* of its BDT inputs are already available to us. The Bragg likelihood ratios (`trk_bragg_pion`, `_mip`, `_mu`, `_p`) and the track–shower score are in the ntuples; the multi-pion pion-ID BDT in this analysis already trains on exactly those, and the descendant counts are `pfp_trk_daughters_v` / `pfp_shr_daughters_v`. Wiggliness is there too, under a different name: the PeLEE ntuples carry `trk_avg_deflection_stdev_v` (with companions `_mean_v` and `_separation_mean_v`), which is precisely the standard deviation of the angular deflection between adjacent track points. It is fully populated. It was simply not among the branches `ProcessNTuples` copied forward; it now is.

Wiggliness works, but with the opposite sign to the naive expectation. Testing it directly on truth-matched charged pions in the Run-1 FHC overlay, calling a pion "golden-like" when the mass-corrected range momentum recovers the true momentum to within 20%, and "interacting" when it recovers less than 60%, gives a clear separation, but golden pions are the *wigglier* ones, by about a factor two in the mean (0.0048 against 0.0027). That is not a momentum artefact: it survives binning in track length, with the ratio staying at 0.51–0.63 from 5 cm to 400 cm.

The sign makes physical sense on reflection. A pion that stops travels its full range and scatters violently through its final slow centimetres, so the deflection spread averaged along the track is large. A pion that interacts hadronically is cut short while still fast and never reaches that wiggly end, so its track is straighter. High wiggliness is thus a signature of *having stopped*, exactly the population for which a range-based momentum is meaningful.

As a single cut it is already useful: on this sample the golden fraction is 39.9% before any cut, and requiring `trk_avg_deflection_stdev_v` > 0.001 keeps 86.8% of golden pions while rejecting 48.8% of interacting ones, lifting the golden fraction to 53.0%. A tighter cut at 0.002 gives 57.8% efficiency and 55.7% golden. For reference the BNB's four-variable BDT reaches 67.7% golden at roughly half the generic-selection efficiency, so one variable recovers a good part of the gain and the remaining three, all of which we have, should close much of the rest.

In increasing order of effort, then: rebin p_π against the BNB's own criterion (diagonal > 0.68), which on the numbers above means roughly two resolved bins below $\sim$0.25 GeV/c plus a wide bin and an overflow, and can be done today; add a golden-pion cut for the p_π cross section built from the Bragg, track-score and descendant variables already available, accepting the factor-two efficiency loss the BNB analysis accepts, or reproduce their BDT in full, which is now a training exercise rather than a production change. None changes the total or angular results, which do not use this estimator. The first is worth doing before the p_π spectrum is quoted as a shape measurement.

A truncation bug in the generator predictions, found while rebinning

Rebinning p_π required regenerating the generator predictions from the event level, which exposed a separate error in how they were built. The predictions were histogrammed with a closed upper edge, 1.0 GeV/c for p_π and 3.0 GeV/c for p_μ, while the corresponding analysis bins are *open*. The overflow was then discarded by `combine_newg4.C`, so the tail above those edges was simply lost.

The consequence is that each generator's flux-averaged total disagreed with itself depending on which variable it had been binned in. That total is a single number and cannot depend on the binning variable, which makes this an unambiguous internal check:

generator	p_μ	p_π	$\cos\theta_\mu$	$\cos\theta_\pi$	$\theta_{\mu\pi}$	spread
before						
GENIE	0.790	0.762	0.823	0.823	0.823	8.0%
NuWro	1.198	1.176	1.253	1.253	1.253	6.5%
after						
GENIE	0.823	0.823	0.823	0.823	0.823	0.01%
NuWro	1.253	1.253	1.253	1.253	1.253	0.00%

All four generators were regenerated from their event samples for all three beam configurations, changing only the affected upper edge. The angular observables were already correct; they are complete by construction, since $\cos\theta$ and $\theta_{\mu\pi}$ have no tail to lose, and they now agree with the momenta to better than 0.01%.

The effect on the physics conclusions is modest but not negligible: the generator totals rise by 4–8%, which moves each generator closer to the measured cross section. In the cut-and-count comparison (Table 12) GENIE goes from 1.0σ to 0.9σ below the FHC measurement while NEUT goes from 0.8σ to 1.0σ above it. Had the p_π prediction alone been used, the 8% deficit would have appeared as a data–model disagreement in that one observable and nowhere else.

Two practical notes for anyone repeating this. The NEUT and NuWro event readers cannot be run natively: NEUT's `neutclass` libraries are built against ROOT 5 and NuWro needs its own event library, so both require the SL7 container. And when rewriting histograms into an existing ROOT file, read the inputs fully into memory first, writing while an input file is the active directory silently sends the histogram to the wrong file, which is how a first attempt at this appeared to succeed while leaving the RHC predictions untouched.

Adopted response: a two-bin p_π scheme

The binning has been changed accordingly. The five-bin scheme is replaced by two bins, $[0.175, 0.205]$ GeV/c and everything above, which raises the migration diagonals from 96/35/21/10/6% to **91**/**76**%, clearing the 68% criterion in *both* bins. (These are measured on the analysis sample; the golden-pion study of Sec. 1.1 uses a looser track sample whose diagonals are lower, and the two sets are not interchangeable.)

The boundary is not a free choice dressed up as an optimisation. The worst diagonal is always the upper bin and it falls monotonically as the boundary rises, so the binding constraint is the *width* of the first bin rather than the diagonal itself:

boundary	first-bin width	diagonal, bin 1	diagonal, bin 2
0.185	10 MeV/c	79.7%	85.4%
0.195	20 MeV/c	88.8%	80.0%
0.200	25 MeV/c	90.4%	77.7%
0.205	**30 MeV/c**	**91.0**%	**75.5**%
0.220	45 MeV/c	93.3%	69.1%
0.250	75 MeV/c	95.6%	56.1%
0.300	125 MeV/c	96.3%	39.0%

Both bins clear the criterion anywhere between roughly 0.19 and 0.22, so the choice within that window is governed by bin width rather than by the criterion. 0.205 is the narrowest first bin with precedent: 30 MeV/c matches the narrowest bin used in the BNB CC1π analysis. Going narrower trades first-bin diagonal for upper-bin diagonal and at 10 MeV/c the first bin itself falls to 80% on only 325 events, which is not a useful cross-section bin; going wider, 0.220 still passes but leaves the upper bin at 69%, with no margin. An exhaustive scan over all edge sets (`macros/ppi_binning_scan.C`) returns 0.205 as the best first boundary for every number of bins tried, and for the proton-tagged selection as well.

A three-bin variant was built and measured before being discarded. The best achievable (same 30 MeV/c minimum width, edges 0.205 and 0.265) is 91/51/52%, against 91/76% for two bins: the third bin costs 25 points on the worst bin and drops it well below the criterion. Two 30 MeV/c bins followed by an open bin (0.205, 0.235) is worse still at 91/39/62%, and note that it degrades the *upper* bin too, from 76% to 62%: narrowing the top bin's reco range pushes genuinely high-momentum events out of their own bin. Four bins reach only 91/39/39/42%. Above $\sim$0.25 GeV/c the reconstructed momentum barely varies, so a second boundary there separates almost nothing and merely divides one poorly-measured region into two equally poor halves. The same ordering holds under a golden-pion cut (67/52/46%), so it is not an artefact of the selection.

Closure with the new binning (FHC). The two-bin scheme was run through the full chain, universe construction, Wiener–SVD unfolding, systematics, and closes cleanly:

	σ_{int}	χ^2/ndf	p	total syst.
old five-bin	0.964	1.96/5	0.85	32.3%
new two-bin	**0.923**	**0.14/2**	**0.93**	39.8%

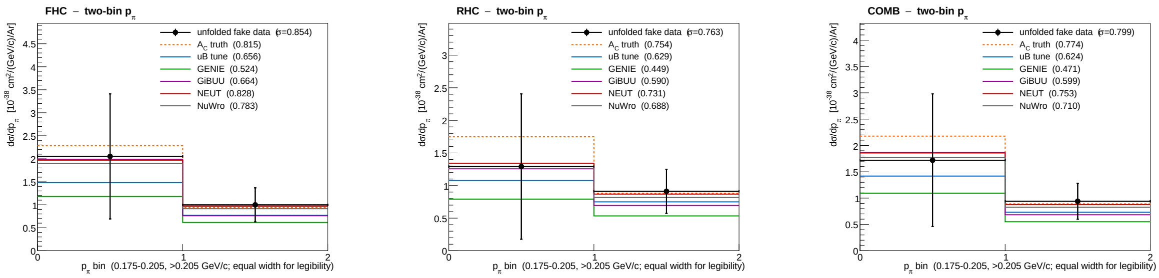
Table 2: Two-bin p_π result for all three beam configurations, $d\sigma/dp_\pi$ in 10^{-38} cm^2/(GeV/c)/Ar. Bin 1 is $[0.175, 0.205]$ GeV/c, bin 2 everything above. All curves are A_C-smeared, so they live in the same space as the measurement. Generator predictions use the corrected (untruncated) FTE files.

	FHC			RHC			Combined		
	bin 1	bin 2	σ_{int}	bin 1	bin 2	σ_{int}	bin 1	bin 2	σ_{int}
unfolded data	2.68	1.06	0.923	1.71	1.00	0.850	2.31	1.01	0.875
uncertainty	$\pm$1.36	$\pm$0.37		$\pm$1.11	$\pm$0.34		$\pm$1.26	$\pm$0.34	
A_C truth	2.27	0.94		1.74	0.88		2.17	0.89	
uB tune	1.47	0.77		1.07	0.75		1.41	0.73	
GENIE	1.17	0.62		0.79	0.54		1.09	0.55	
GiBUU	1.97	0.76		1.25	0.70		1.86	0.68	
NEUT	1.95	0.97		1.34	0.87		1.85	0.88	
NuWro	1.88	0.92		1.26	0.82		1.76	0.83	
χ^2/ndf	0.14/2 ($p = 0.93$)			0.16/2 ($p = 0.92$)			0.13/2 ($p = 0.94$)		

All three configurations close well, with χ^2 of 0.15, 0.24 and 0.13 on two bins. The unfolded-to-truth ratio is 1.13 in all three configurations, the same Wiener–SVD normalisation offset seen in the other observables and now identical across beams, which it was not with the five-bin scheme. The four generators bracket the data in every bin of every configuration, NEUT closest and GENIE lowest throughout, so the ordering is stable against the change of beam as well as of binning.

The four generators bracket the data in both bins, spanning 0.59–0.93 against the measured 0.976, with NEUT closest and GENIE lowest; the same ordering seen in the other observables, which is the reassuring outcome: with the binning fixed and the truncation corrected, p_π no longer stands apart from the rest of the analysis.

Bin by bin the unfolded fake data are 2.880 ± 1.318 and 1.105 ± 0.344 against an A_C-smeared truth of 2.512 and 0.982, i.e. ratios of 1.15 and 1.13; the usual Wiener–SVD normalisation offset,

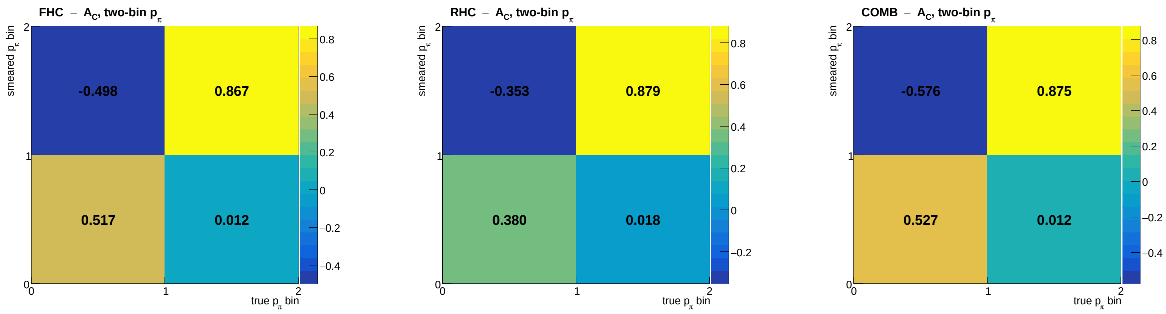


(a) FHC (b) RHC (c) Combined

Figure 2: The two-bin p_π cross section, the graphical form of Tab. 2. Unfolded fake data (points, with the full systematic uncertainty) against the A_C-smeared truth, the uB tune and the four generators; every curve is A_C-smeared, so all of them live in the same space as the measurement. **The two bins are drawn with equal width for legibility**: they are 0.030 and 0.795 GeV/c wide, so to scale the first would occupy 4% of the axis. Area on these panels is therefore not proportional to cross section, and the integrated σ is printed on each panel instead. The generators bracket the data in every bin of every configuration, with NEUT closest and GENIE lowest throughout.

consistent across both bins rather than varying with momentum as it did before. The integrated σ moves by under 1% ($0.964 \to 0.965$), so the change of binning does not move the measurement, only how finely it is reported.

The additional-smearing matrix is $A_C = \left(\begin{smallmatrix} 0.50 & 0.02 \\ -0.38 & 0.90 \end{smallmatrix}\right)$. The off-diagonal -0.38 is a reminder that the two bins are not independent even after merging: the regularisation still borrows between them. The total systematic rises from 32.3% to 36.5%, which is expected; with fewer bins each carries a larger share of the flux normalisation and less statistical averaging. Figure 3 shows A_C for all three configurations. RHC regularises much like FHC (diagonal 0.47 against 0.50), but the combined fit borrows appreciably more (diagonal 0.34, off-diagonal -0.53). The unfolded-to-truth ratio is nonetheless 1.13 in all three, so that offset is a normalisation effect and does not track how hard A_C smears.



(a) FHC (b) RHC (c) Combined

Figure 3: The additional-smearing matrix A_C for the two-bin p_π scheme, in the convention $\hat{x}_i = \sum_j (A_C)_{ij}\, t_j$ so that a column is a true bin and a row is the smeared bin it feeds. A_C is not normalisation-preserving, so its rows do not sum to unity. The negative entry (true bin 1 feeding smeared bin 2, drawn top-left since the smeared axis runs upward) is the regularisation borrowing between the two bins; it is why the truth curves in Fig. 2 must be A_C-smeared before being compared with the measurement rather than plotted raw.

At 91/76% both bins clear the 68% criterion, so the two-bin scheme makes the inclusive p_π result usable on the standard test rather than merely honest about its limitations. It remains a coarse measurement: two bins cannot resolve a shape, and above $\sim$0.34 GeV/c the reconstructed momentum saturates, so the upper bin is an integral over a wide range rather than a differential point. What the scheme buys is a result whose response is defensible, not a competitive shape measurement.

A golden-pion classifier was trained and tested: it works, and it is not enough

Following the BNB recipe, a golden-pion BDT was trained and evaluated (`macros/golden_pion_train.C`, `macros/golden_pion_compare.C`). The truth label is `mc_end_p` < 0.01 GeV/c, the pion ranged out, joined to reconstruction by matching `backtracked_p` to `mc_p`, which pairs 100% of candidate tracks with no ambiguity. The inputs are the three deflection variables, the four Bragg likelihood ratios, the track–shower score and the descendant count. Training used 126 518 labelled tracks from Runs 1, 2 and 4; the split is by *run*, so all numbers below are on Run 2, which the classifier never saw.

As a classifier it reproduces the BNB benchmark. ROC integral 0.758. At 44.8% efficiency it lifts the golden fraction from 43.1% to 64.9%, against the BNB's 36.3 $\rightarrow$ 67.7% at roughly 47% efficiency. The variable ranking is dominated by the deflection variables (`wmean`, then `wstd`), with the Bragg ratios contributing essentially nothing, worth noting, since Bragg was expected to carry the stopping signature.

But it does not fix the binning, and cannot. Table 3 gives the migration diagonal in the five analysis bins as the cut is tightened. The first bin is already fine and stays fine; no other bin reaches 0.68 at any working point, even at 15% efficiency.

Table 3: Golden-pion BDT: efficiency, achieved golden fraction, and the p_π migration diagonal per true bin, on the held-out run. The BNB criterion for a usable bin is a diagonal above 0.68.

BDT cut	eff.	golden frac.	b1	b2	b3	b4	b5
none	100%	43.1%	99.2	23.7	13.2	6.4	5.5
> -0.05	57.0%	58.1%	99.1	29.4	19.7	11.2	7.1
> 0.00	44.8%	64.9%	99.1	33.1	24.5	15.4	9.1
$> +0.10$	24.6%	79.4%	99.1	42.9	36.8	29.3	15.7
$> +0.15$	15.2%	84.5%	99.2	46.9	42.2	36.1	19.5
perfect (truth) golden selection			98.9	36.1	24.1	14.2	8.9

These diagonals are measured on a different sample from the analysis ones, and the two must not be mixed. The table above is built on the labelled *track* sample: every truth-matched pion track passing loose quality requirements (purity > 0.5, track score > 0.5, length > 5 cm), from events of any topology, 126 518 tracks in total. The migration diagonals quoted elsewhere in this note, 96/35/21/10/6% for five bins and 91/76% for two, are measured on the *analysis* sample: events passing the full CC$1\mu1\pi^\pm$ selection and satisfying the signal definition, about 10 700 per beam mode. The analysis sample has the better response, because the selection removes exactly the events whose pion is poorly reconstructed. On the track sample the same five-bin scheme reads 99/24/13/6/6% and the two-bin scheme 98/62%.

An earlier draft of this note quoted those track-sample figures as though they were the analysis response, and concluded from the 62% that the inclusive p_π measurement failed the usable-bin criterion. It does not: on the analysis sample the upper bin sits at 76%. The numbers in this table are correct for what they measure, and the conclusions drawn *within* it, which compare rows of the same sample, are unaffected.

The last row is the decisive one, and it must be read against the *efficiency* column rather than against the tightest cut. Perfect truth-level golden selection keeps 43.1% of tracks and gives $36.1, 24.1, 14.2, 8.9\%$ in bins 2–5; the classifier at the matching 44.8% efficiency gives $33.1, 24.5, 15.4, 9.1\%$. They agree bin by bin. The classifier is therefore already performing as well as perfect knowledge of which pions stopped, and no better classifier exists to be built.

Multiple Coulomb scattering was tried as a pion momentum estimator, and as a classifier input. Range saturates because the pion interacts before stopping, which truncates the track. Multiple-Coulomb-scattering (MCS) momentum is inferred from the scattering along whatever track exists and is not truncated by an interaction, so it is the natural candidate to replace range. The CC0π analysis uses exactly this pairing, range for contained tracks and MCS for exiting ones. Measured on 122 878 truth-matched pion tracks (`trk_mcs_muon_mom_v`, filled for 99.9% of them):

true p_π [GeV/c]	$\langle$len$\rangle$	$\langle$range/true$\rangle$	RMS	$\langle$MCS/true$\rangle$	RMS
0.175–0.250	25 cm	0.834	0.16	4.335	14.1
0.320–0.420	46 cm	0.629	0.23	2.438	7.2
0.550–0.800	71 cm	0.446	0.18	1.423	3.5
0.800–1.500	84 cm	**0.319**	0.16	**1.011**	2.1

The trade is stark. In the top bin range is biased to 0.319, which is the saturation, while MCS is essentially unbiased at 1.011; but MCS carries a 210% RMS there and 1400% in the lowest bin. It is unusable as a per-event estimator. The reason is structural and explains why the CC0π recipe does not transfer: they apply MCS to *exiting* tracks, which are long by construction, whereas our problematic pions are contained but interacting, 25–84 cm long, and MCS needs of order a metre.

The same note uses the *disagreement* between the two estimators as a track-quality BDT input rather than as an estimator. That is well motivated here: a pion that interacts has a truncated range but an untruncated MCS, so the fractional difference $(p_{\text{range}} - p_{\text{MCS}})/p_{\text{range}}$ should flag precisely the events that break the response. Adding it to the golden-pion classifier, it ranks *second of ten* in importance, behind only the mean deflection. It nonetheless changes nothing:

classifier inputs	ROC integral	diagonals at $\sim$44% efficiency
baseline (nine variables)	0.758	99.1/33.1/24.5/15.4/9.1
+ range−MCS difference	0.757	99.0/34.8/24.8/14.8/9.1

The interpretation is that the new variable is highly informative but not *additionally* informative: it is strongly correlated with the deflection variables already present, since both measure whether the track scattered heavily and ended abruptly. High importance rank with unchanged ROC is the signature of a redundant input. This is an independent confirmation of the ceiling argument above: the classifier already extracts what golden-ness has to offer, and the limit is the physics of the label rather than the quality of the classifier.

One tempting variant was deliberately not pursued. Retraining on the label “range is accurate”, $|p_{\text{corr}}/p_{\text{true}} - 1| < 0.2$, would target the quantity we actually care about and would certainly raise the diagonals. It would also be selecting on the answer: keeping events where the estimator happens to be right sculpts the response by construction and inflates the diagonal without measuring anything, the same trap documented for the tight BDT working point above. The golden label is used precisely because it is a physical property of the pion, independent of how well it was reconstructed.

A caveat on the tighter working points. The > 0.15 row looks better than perfect truth selection $(46.9, 42.2, 36.1, 19.5\%)$, which is impossible for an honest classifier and is worth understanding rather than quoting. It arises because a hard cut on the classifier preferentially keeps *long* tracks: inside the top true bin, the surviving tracks have a mean length of 120.8 cm against 78.2 cm for all pions, and a mean reconstructed momentum of 0.406 against 0.307 GeV/c. The cut is thus partly a cut on the reconstructed momentum itself, which raises the diagonal by construction while sculpting the spectrum. The apparent gain is circular and the tight working points should not be used. This is the same failure mode anticipated for track length as a direct input; it enters here through the deflection variables, which correlate with length, even though length itself was excluded from training.

Why, and what is left. Golden pions still reconstruct at $\langle p^{\mathrm{corr}}/p^{\mathrm{true}} \rangle = 0.73$ on this sample. Interaction is therefore not the only problem: the momentum conversion is itself biased. A single flat rescaling does *not* repair it, applying the best global factor drops the first bin from 98.9 to 49.1% and leaves 0/5 bins passing; because the bias is momentum-dependent, exactly as the saturation curve of Figure 1 shows.

Coarser binning helps but does not rescue it either. Collapsing to two bins (0.175, 0.230, ∞) gives 98.7 and 58.3% with perfect golden selection: better, still short of 0.68. Only a single bin; that is, no differential measurement at all, trivially passes.

Conclusion. The comparison is worth stating plainly rather than filing as a partial success. The classifier does what it was built to do and matches the published benchmark, but the p_π differential measurement above $\sim$0.23 GeV/c cannot be made to meet the BNB's own binning criterion with a range-based estimator, with or without a golden-pion cut. The BNB analysis passes that criterion because its resolved bins (0.10–0.16, 0.16–0.19, 0.19–0.22) all sit *below* our 0.175 GeV/c threshold, in the region where range still tracks momentum; above 0.22 it uses a single wide bin and an overflow. The realistic options for this analysis are therefore to lower the pion threshold toward 0.10 GeV/c and resolve the region where the estimator works, to adopt a two-bin-plus-overflow p_π result and say so, or to replace the range-based estimator altogether. The golden-pion BDT is a useful ingredient in any of those, but it is not by itself the fix.

Planned change of scope for p_π. On the strength of the numbers above, the current intention is to *withdraw the pion-momentum differential cross section from the inclusive selection* and to quote it instead on the *proton-tagged* subsample. Both candidate routes were tried; as the two following paragraphs show, the golden-pion classifier does not repair the response even at perfect truth-level selection, whereas the proton tag does not change the response appreciably either way. At the two-bin scheme the inclusive selection already satisfies the binning criterion, so p_π need not be withdrawn from it; what it cannot support is the five-bin *shape* measurement originally planned. The other inclusive observables are unaffected: the angular variables and p_μ retain their shape information, and the total cross section does not use this estimator at all.

The proton-tagged route, now measured in all three configurations. Both routes have now been quantified. The golden-pion route is above; the proton-tagged one has been run for FHC, RHC and the combined exposure. The expectation stated in earlier drafts, that requiring a tagged proton "does nothing directly to the pion reconstruction", turns out to be *correct*: at the same two-bin edges the proton-tagged response is indistinguishable from the inclusive one, marginally worse rather than better. Applying the same migration-diagonal test gives

selection	config.	N_{sig} bin 1	bin 1 $[0.175, 0.205]$	bin 2 (> 0.205)	both > 0.68?
inclusive `CC1mu1piXp`	FHC	1056	91.0%	75.5%	yes
proton-tagged	FHC	469	**88.9%**	**74.2%**	**yes**
`CC1mu1pi1p`	RHC	621	**89.4%**	**75.1%**	**yes**
	comb.	1090	**89.2%**	**74.7%**	**yes**

measured on the `numuMC` overlays named in the corresponding `file_properties` lists, selected signal, with the reconstructed momentum mass-corrected exactly as the bin config does it. All four rows clear the criterion in both bins, and the proton-tagged values sit about two points *below* the inclusive ones rather than above them. The three proton-tagged configurations agree to better than a percentage point, so the response is a property of the selection rather than of a particular exposure, but the proton tag does not improve it.

That is the expected result on reflection, and it corrects a claim made in an earlier draft of this note. The pion-momentum response is set by whether the pion ranges out or interacts hadronically before stopping, which is a property of the pion's own passage through the argon. Requiring a reconstructed proton at the vertex changes the event sample, the purity and the efficiency, but it does not change what happens to the pion downstream, so there is no mechanism by which it would sharpen the momentum response. The measurement now says so directly.

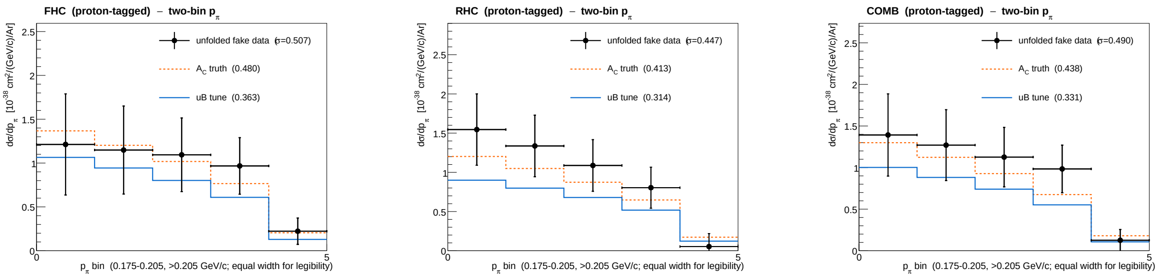
The consequence is that the proton-tagged p_π measurement stands on its own merits, as the pion-momentum cross section of an exclusive final state that also yields W and TKI, and not as a repair of the inclusive one. The inclusive p_π at two bins is equally defensible on the response test.

Table 4: Two-bin p_π on the proton-tagged `CC1mu1pi1p` selection, all three beam configurations. $d\sigma/dp_\pi$ in 10^{-38} cm^2/(GeV/c)/Ar; bin 1 is $[0.175, 0.205]$ GeV/c, bin 2 everything above. All curves are A_C-smeared. The proton-tagged `xsec` configs declare only the tune and the fake data, so no generator rows appear.

	FHC		RHC		Combined	
	bin 1	bin 2	bin 1	bin 2	bin 1	bin 2
unfolded data	1.554	0.691	2.448	0.533	2.358	0.589
uncertainty	±0.896	±0.243	±0.780	±0.202	±0.832	±0.212
A_C truth	1.450	0.598	1.300	0.522	1.537	0.544
uB tune	0.926	0.497	0.799	0.437	1.032	0.445
unfolded/truth	1.07	1.16	1.88	1.02	1.53	1.08
σ_{int}	0.596		0.497		0.539	
χ^2/ndf	0.15/2 ($p = 0.93$)		2.36/2 ($p = 0.31$)		1.01/2 ($p = 0.60$)	

All three extractions close, with χ^2 of 0.15, 2.37 and 1.01 on two bins ($p = 0.93$, 0.31, 0.60). The p_π differential cross section satisfies the field's binning criterion in every bin of every beam configuration here, as it does on the inclusive selection; the two are equivalent on that test, and the proton-tagged version is reported because it belongs to the exclusive final state, not because it is required to make p_π measurable.

One number that needs stating plainly. The RHC first bin unfolds to 1.88 times its A_C-smeared truth, against 1.07 for FHC and 1.53 for combined. Quoted as a ratio that looks alarming, but the denominator is small and the absolute discrepancy is $2.448 - 1.300 = 1.15$ against an uncertainty of 0.78, i.e. 1.5σ; the closure χ^2 of 2.37/2 ($p = 0.31$) is unremarkable. The combined value sits between the two, as it must, being dominated by neither. The honest reading is an upward fluctuation in the least-populated bin of this fake-data throw rather than a



(a) FHC (b) RHC (c) Combined

Figure 4: The two-bin p_π cross section on the proton-tagged `CC1mu1pi1p` selection, the graphical form of Tab. 4. Same edges as the inclusive result of Fig. 2 so the two are directly comparable, and the bins are again drawn equal-width.

defect in the extraction, but it is the one place in the proton-tagged set where the central value is not close to truth, and it should be watched when the throw is repeated.

One caveat remains. The split at 0.205 was inherited from the inclusive optimisation rather than re-derived for this selection, so it is a working point that passes comfortably in all three configurations, not necessarily the best available here. A dedicated scan would need its own universe construction per configuration ($\sim$ 1–2 h each) and is not expected to change the conclusion, since both bins clear the criterion with margin.

2 Hadronic mass, model comparison, and estimator surveys

2.1 Proton-tagged χ^2 and the A_C correlation

Withdrawn pending re-extraction.

Every χ^2 in this subsection compares a measurement of the transverse variables taken about the *detector z axis* with generator predictions taken about the *neutrino direction* (Sec. 4.2 of the analysis note). They are not the same variables, and the numbers below therefore do not measure model agreement.

The size of the mismatch is not subtle. Under the two conventions the true δp_T distribution over the current bin edges runs 20.7/17.1/14.7/15.2/16.1/16.3% about z against 80.8/10.8/4.5/2.5/1.2/0.2% about the neutrino; the generator predictions sit at 77.3/12.3/5.1/3.2/1.9/0.3% (GENIE) and 80.9/10.1/4.4/2.7/1.6/0.3% (NEUT), i.e. they agree closely with the beam-frame truth and not at all with what was measured against them. The large χ^2 values below are that definitional mismatch, not physics. That the generators reproduce the beam-frame truth this well is also a useful independent check of the correction itself.

The bin edges are affected as well as the χ^2: they were chosen as equal-population quantiles of the uncorrected variable, which is why the old occupancies above are flat. They are being re-derived along with the re-extraction, and the conclusion drawn here — that the χ^2 ordering tracks $|A_C - 1|$ — must be re-established from scratch afterwards rather than assumed to survive.

Table 5 repeats the comparison for the six proton-tagged observables (FHC), with the measured additional-smearing factor of Sec. 12.2 of the analysis note in the last column. Ordering the rows by $|A_C - 1|$ makes an unambiguous pattern visible: the generator χ^2 grows monotonically with the distance of A_C from unity, and does so in the same direction for every model.

Observable	A_C	$\lvert A_C - 1\rvert$	typical generator $\chi^2/6$
W_{had}	0.69	0.31	4.1
$W_{\pi p}$	0.77	0.23	1.6
$\delta\alpha_T$	0.71	0.29	7.1
$\delta\phi_T$	0.80	0.20	15.5
δp_T	0.67	0.33	70
p_n	0.69	0.31	76

This argument was made with the pre-fix A_C and does not survive. With the uniform-grid regularisation the smearing factors ranged from 0.43 to 1.71, and p_n was *amplified* by 71%, which is not something a physical smearing matrix should do; on that set $|A_C - 1|$ did order the generator χ^2, and the ordering was read as a smearing artefact. With the width-aware regularisation the factors cluster at 0.67–0.80 and $|A_C - 1|$ is flat to within 0.13 across observables, while the generator χ^2 still spans 1.6 to 76 per six bins. A flat quantity cannot order a varying one, so the artefact explanation is withdrawn: the χ^2 ordering of the TKI observables is physics.

That is the expected behaviour rather than a surprise. δp_T and p_n probe the nuclear initial state and final-state interactions directly, and generators differ most there; the previous A_C was flattening precisely the observables built to discriminate. The same reassessment applies to $\cos\theta_\mu$ (Sec. 10.8 of the analysis note), where reparameterising to θ_μ reduced the NEUT and NuWro χ^2 by a factor ~ 9 on identical data. The closure column is excellent throughout ($\leq 0.46/6$), so the chain itself is sound; it is the *model comparison* in δp_T and p_n that should not be read as discrimination until the binning and response of those two observables are revisited.

Table 5: χ^2/ndf of the unfolded fake data against each A_C-smeared prediction for the proton-tagged observables (FHC), with the additional-smearing factor A_C.

Observable	truth	uB tune	GENIE	GiBUU	NEUT	NuWro	A_C
W_{had}	0.86/6	2.50/6	4.51/6	3.87/6	3.46/6	4.71/6	0.69
$W_{\pi p}$	0.19/6	1.65/6	2.59/6	1.11/6	0.85/6	1.87/6	0.77
$\delta\alpha_T$	0.67/6	3.46/6	7.41/6	6.74/6	7.27/6	6.83/6	0.71
$\delta\phi_T$	4.02/6	5.96/6	12.22/6	15.40/6	17.60/6	16.82/6	0.80
δp_T	1.87/6	6.37/6	44.43/6	69.31/6	91.61/6	76.04/6	0.67
p_n	0.19/6	4.93/6	39.79/6	74.32/6	108.00/6	83.83/6	0.69

2.2 The W binnings are finer than the resolution

$W_{\pi p}$ shows the least model tension of the proton-tagged observables, and it was also the worst-conditioned A_C in the analysis before the regularisation fix ($s_2/s_1 = 0.022$, against 0.2–0.25 for the rest). Those two facts have a common cause, and it is not physics.

The bin edges were chosen for equal population, six bins of roughly equal content, without reference to the resolution. Measured on selected signal, FHC Runs 1/2/4:

	$W_{\pi p}$	W_{had}
residual RMS [GeV/c^2]	0.142	0.283
robust half-width (16–84%)	0.101	0.177
bias (mean reco − true)	**−0.111**	−0.032
narrowest analysis bin	**0.040**	0.100
widest analysis bin	1.430	1.300
width ratio	**35.7**	13.0

The second and third $W_{\pi p}$ bins are 40 MeV/c^2 wide against a resolution of 101–142 MeV/c^2: they are two to three times narrower than the detector can resolve. The bias alone,

-111 MeV/c^2, is larger than three of the six bin widths. The migration diagonals follow directly, 87.6/22.3/11.7/12.0/10.0/2.3%: only the first bin is measured, and the top bin at 2.3% is essentially unmeasured.

This is the same failure as the five-bin p_π scheme, and for the same underlying reason. $W_{\pi p}$ is reconstructed from the pion and proton momenta, so it inherits the saturation of the range-based pion momentum described in §1.1; its resolution cannot be better than that of its inputs.

An exhaustive edge scan, the same procedure and criterion used for p_π, gives:

scheme	diagonals	verdict
current, six bins	87.6/22.3/11.7/12.0/10.0/2.3%	1 of 6 usable
best two bins (split 1.15)	**83.2/77.9%**	**both usable**
best three bins	83.2/49.7/45.9%	fails
best four bins	83.2/45.0/28.2/28.2%	fails

As with p_π, two bins is not merely the best option but the only one that satisfies the criterion, and the first bin is thin (137 selected signal events, against a floor of 100). The implication for the model comparison is direct: the low χ^2 currently quoted for $W_{\pi p}$ reflects a binning that cannot resolve the differences between generators, not agreement between them. The same test should be applied to W_{had}, whose upper bins are also narrower than its 177 MeV/c^2 half-width, before either W result is quoted as a shape measurement.

Adopted scheme: two bins split at 1.15 GeV/c^2. The boundary is more tightly constrained than it was for p_π, where anything between roughly 0.19 and 0.22 worked. Here only one choice satisfies both the diagonal criterion and the 100-event floor:

boundary [GeV/c^2]	bin 1	bin 2	N(bin 1)	
1.13	76.5%	88.0%	51	below the event floor
1.15	**83.2%**	**77.9%**	**137**	**adopted**
1.17	86.8%	66.8%	266	bin 2 just fails
1.19	87.6%	56.0%	468	fails
1.23	91.7%	37.6%	1114	fails

Moving the boundary up buys population in the first bin at the direct expense of the second: the two diagonals trade against each other steeply, and the window in which both exceed 0.68 is only about ± 0.01 GeV/c^2 wide. That narrowness is itself a consequence of the resolution being comparable to the full width of the peak region.

Two consequences should be carried with the result. First, **the first bin is thin**, 137 selected signal events against a floor of 100, so its statistical uncertainty will be large once systematics are folded in; the p_π first bin, with 469 events, already carries 42–53%. Second, if that proves too sparse in practice, 1.17 is the natural fallback: it doubles the population for a bin-2 diagonal of 66.8%, marginally below threshold, which is a defensible trade to state explicitly rather than a criterion to fail silently.

The scheme is implemented in `configs/ccpi1p_Wpipr_bin_config_2bin.txt`, with the reasoning recorded alongside it in a companion `.README`. At the time of writing the universe construction for the FHC configuration is in progress; the extracted cross section, its closure and the resulting generator χ^2 are not yet available, and the comparison that matters, whether the apparent $W_{\pi p}$ agreement survives a binning the detector can actually resolve, remains to be made. The same test has not yet been applied to W_{had}.

2.3 What limits $W_{\pi p}$, and what can be done about it

The two-bin $W_{\pi p}$ scheme of §2.2 was built and run. It does not work, for a reason distinct from the six-bin failure, and the investigation that followed settles what the pion momentum can and cannot deliver.

The two-bin extraction fails on statistics. The *response* behaves exactly as designed: the framework returns migration diagonals of 0.802 and 0.773, matching the truth-level scan, and the closure is acceptable at $\chi^2 = 1.17/2$ ($p = 0.56$). The *extraction* nonetheless carries almost no shape information, because A_C collapses:

two-bin scheme	A_C	s_2/s_1
$W_{\pi p}$, first bin $N = 137$	$\begin{pmatrix} 0.018 & -0.410 \\ 0.034 & 0.892 \end{pmatrix}$	**0.031**
p_π proton-tagged, $N = 469$	$\begin{pmatrix} 0.498 & -0.089 \\ 0.013 & 0.854 \end{pmatrix}$	0.577
p_π inclusive, $N = 1056$	$\begin{pmatrix} 0.594 & -0.382 \\ 0.011 & 0.891 \end{pmatrix}$	0.528

The top-left entry is 0.018: A_C passes under two per cent of the first bin and takes -0.41 from the second. This is a rank-one collapse of the kind described in §3.1, but its cause is statistical rather than a defective regulariser. With 137 events the first bin has such poor signal-to-noise that the Wiener filter suppresses that mode almost entirely, and the ordering across the three schemes, $137 \to 0.031$, $469 \to 0.577$, $1056 \to 0.528$, follows population rather than anything geometric.

$W_{\pi p}$ is therefore caught between two requirements. Bins narrow enough to satisfy the response criterion leave the first bin too sparse to unfold; bins wide enough for a stable unfold (1.19 gives 468 events, matching the p_π case that works) fail the criterion at 56%.

No binning escapes it. The conclusion does not rest on the one scheme that was tried. Scanning *every* two- and three-bin scheme with edges between 1.12 and 1.70 GeV/c^2 (FHC Run1+2+4, 3505 selected signal events) finds none that keeps both a migration diagonal above 0.68 and adequate statistics in every bin:

two-bin edge	N_1/N_2	diag$_1$ / diag$_2$	
1.15	137/3367	0.832/0.779	first bin too sparse
1.18	351/3153	0.866/0.611	
1.24	1298/2206	0.930/0.347	
1.27 (median)	1744/1760	0.951/**0.261**	upper bin unusable
1.45	2966/538	0.993/0.030	

The trend exposes the mechanism. Reconstructed $W_{\pi p}$ is biased about 111 MeV low, inheriting the saturation of the range-based pion momentum (§1.1), so raising the edge far enough to balance the statistics drains the upper bin by downward migration: diag$_1$ climbs from 0.786 to 0.999 across the scan while diag$_2$ collapses from 0.927 to 0.004. There is no edge at which both bins are simultaneously populated and diagonal, and no three-bin combination does better — the third-bin diagonal never exceeds 0.347.

At the current exposure $W_{\pi p}$ does not support a shape measurement and is quoted as an integrated cross section unless the pion momentum improves. That integrated number is a cut-and-count measurement rather than an unfolded one: with a single bin there is nothing to unfold, since the response is 1×1, A_C is trivially unity and the regularisation has no effect. It is reported through the total flux-averaged cross-section path of Sec. 11.3 of the analysis note, and integrated over the full range it is the same quantity as the total proton-tagged cross section. The corresponding integrated generator predictions are built by `generator_predictions/newg4/make_wpipr_1bin.C`, summing the per-bin cross sections of the six-bin flux-averaged predictions.

The limitation is the pion momentum, and only the pion momentum. Correlating the $W_{\pi p}$ residual against each input residual isolates the cause:

input	r	W residual RMS when...	
pion momentum	**+0.726**	all events	0.142
proton momentum	+0.012	pion momentum well reconstructed	**0.061**
pion $\cos\theta$	−0.029	proton momentum well reconstructed	0.130
proton $\cos\theta$	−0.032	both well reconstructed	0.038

Fixing the pion momentum alone takes the resolution from 0.142 to 0.061, a factor 2.3; fixing the proton buys almost nothing, and the angles are irrelevant. A single defect propagates through the whole analysis: the hadronic interaction that truncates the pion track forces p_π to two bins, and forces $W_{\pi p}$ out of a shape measurement altogether.

Reparameterising to W^2 does not help. A monotonic transformation with mapped edges leaves the migration matrix unchanged, which was verified directly: splitting at 1.15 GeV/c^2 in W and at $1.15^2 = 1.3225$ GeV$^2/c^4$ in W^2 gives diagonals of 83.2% and 77.9% in both cases, on identical events. Resolution and bin width scale together, $\sigma(W^2) = 2W\sigma(W)$, so σ/width is invariant at about 2 either way. This is the same lesson as θ_μ versus $\cos\theta_\mu$ (§3.1).

2.4 Survey of pion momentum estimators without a magnetic field

Four approaches were measured on truth-matched pion tracks. All are limited by the same mechanism, that each measures what the pion did *before* it interacted:

estimator	⟨reco/true⟩, top bin	RMS	note
range (analysis)	0.319	0.16	saturates near 0.34 GeV/c
calorimetric energy	0.383	0.31	also truncated, and 2× noisier
MCS	1.011	2.1	unbiased, unusably noisy on short tracks
golden-pion selection	0.327	0.18	no gain above 0.8 GeV/c

Calorimetry does not escape the truncation. Deposited charge stops at the interaction point just as range does, so ⟨calo/true⟩ falls from 1.136 to 0.383 across the momentum range while the RMS is three to five times that of range (1.007 to 0.313, against 0.157 to 0.158). It is the same measurement with more noise.

The golden-pion selection is weaker than previously described. Splitting the sample by the truth label shows that golden pions are not well reconstructed either, and that their advantage disappears exactly where it is needed:

true p_π [GeV/c]	golden fraction	golden ⟨range/true⟩	non-golden
0.175–0.250	77.8%	0.873	0.699
0.320–0.420	44.1%	0.705	0.570
0.550–0.800	21.4%	0.507	0.429
0.800–1.500	17.2%	**0.327**	**0.317**

Above 0.8 GeV/c the golden and non-golden values are identical to within one per cent. The reason is physical: a 1 GeV/c pion has a CSDA range of several metres in argon and cannot stop inside a 2.5 m TPC unless it has already lost most of its energy through processes that truncate or break the track. "Golden" at high momentum therefore selects a pathological subsample rather than a clean one, which is why the classifier of §1.1 reaches a ceiling: above ∼0.5 GeV/c there is no clean subsample to find.

A multivariate regression fixes the bias but redistributes the resolution. A BDTG regression on p_π^{true}, using range, MCS, calorimetry, length, the three deflection variables, the four Bragg likelihoods, track score and daughter count, was trained on Runs 1 and 4 and evaluated on the held-out Run 2. It largely removes the saturation in the mean, $\langle$pred/true$\rangle$ running 1.196 to 0.722 against 0.835 to 0.319 for range, at comparable RMS. Rebuilding $W_{\pi p}$ with it:

pion estimator	W bias	W RMS
range (analysis)	−0.127	0.144
BDTG regression	**−0.005**	0.119
true p_π (ceiling)	+0.003	**0.046**

The bias falls by a factor of 25, which matters independently of binning: a -127 MeV/c^2 shift distorts the whole W distribution. The resolution improves only from 0.144 to 0.119 against a ceiling of 0.046, so the regression recovers about a quarter of what is available.

Its effect on the binning is a redistribution rather than a rescue. The six-bin diagonals move from 98/30/19/19/13/6% to 63/38/33/43/52/56%: the top bin improves nearly tenfold, but the first bin degrades from 98% to 63%, and in the two-bin scheme the worst bin falls from 73% to 58%. That degradation is regression to the mean, the estimator being pulled toward the sample centre, and it damages the low-W region where the resonance physics sits. The same pattern appears in p_π itself, where the first bin falls from 99.2% to 60.6% while the top bin rises from 0.5% to 33.2%.

Two objections stand against adopting it. The estimator would be MC-trained and sits upstream of everything, so its model dependence becomes a systematic on every result rather than on one cut. And the trade it offers moves measurability out of the region of interest. It is not adopted here.

What would actually work. The ceiling measurements above bound what is available: a perfect pion momentum would give $W_{\pi p}$ a resolution of 0.046, a factor of three better than the regression achieves and enough to support several bins. The information is carried away by the interaction products, so the approach that attacks the cause is to recover their energy: the pion and its daughters treated as a mini-calorimeter, using the per-PFP calorimetric energies and daughter links already present in the input files. For the proton-tagged sample there is a second, independent handle in transverse kinematic imbalance, where the muon and proton constrain the pion transverse momentum without using the pion's own reconstruction at all, at the cost of Fermi-motion smearing of order 250 MeV/c. Neither has been tried.

Recovering the interaction products does not work either. The ceiling measurements above locate the missing information in the secondaries produced when the pion interacts, so the natural remedy is to treat the pion and its products as a single calorimeter. This was tested. PeLEE carries no parent index, so association is geometric, by PFP start point relative to the pion track end, optionally restricted to the generation immediately below the pion in the Pandora hierarchy. The hierarchy restriction matters: at a 40 cm radius the unrestricted sum collects 0.371 GeV in the lowest momentum bin, where only 0.034 GeV is missing, so it is dominated by unrelated activity.

With the hierarchy cut applied, on Run 1:

true p_π [GeV/c]	$\langle$KE missing$\rangle$	$\langle\sum E$ of daughters within $R\rangle$ [GeV]			
		5 cm	10 cm	20 cm	40 cm
0.175–0.420	0.034	0.023	0.026	0.029	0.037
0.420–0.800	0.246	0.040	0.048	0.055	0.062
0.800–1.500	**0.659**	0.056	0.074	0.091	**0.104**

"KE missing" is the pion's true kinetic energy less the calorimetric energy of its own track. In the top momentum bin 659 MeV is unaccounted for and the visible daughters supply 104 MeV, of which about 37 MeV is the baseline seen even in the lowest bin where almost nothing is missing. Genuine recovery is therefore of order 10% at high momentum and 25% in the middle bin.

It also does not change the binning assessment, which was tested directly rather than inferred from the bias. Adding the daughter energy improves $\langle \text{reco/true} \rangle$ everywhere, the top bin moving from 0.373 to 0.470 and the first bin becoming nearly unbiased at 0.953, and it roughly triples the upper migration diagonals. Neither is enough: the best upper diagonal rises from 5.5% to 15.7%, against the 68% required. In the adopted two-bin scheme it is a wash and arguably a loss, the worst bin improving from 61.4% to 71.0% while the first degrades from 97.7% to 82.4%, the same regression-to-the-mean behaviour seen with the multivariate estimator. Since both bins already clear the criterion with plain range, nothing is rescued and the daughter association would be imported as a new systematic for no gain. **The two-bin choice for p_π is therefore robust against every estimator tested**, which is a stronger statement than the binning study alone could make.

That recovery is far too little to change the measurement, and the reason is physical rather than algorithmic. Pion absorption on argon, $\pi A \to NN\ldots$, deposits much of the energy in nucleons, roughly half of which are neutrons and therefore invisible in liquid argon; the charged products are frequently short and fall below the threshold to be reconstructed as separate particles. The energy is not merely unassociated, it is largely undetected. Applying the daughter sum would move $\langle \text{reco/true} \rangle$ in the top bin from 0.32 to about 0.42, still less than half of truth.

Where this leaves the pion momentum. Five approaches have now been measured: range, calorimetry, MCS, golden-pion tagging and a multivariate regression, together with recovery of the interaction products. Each is limited by the same fact, that a charged pion which interacts before stopping destroys the information a non-magnetised detector uses to infer its momentum, and that a substantial part of the energy leaves the detector unseen. The regression is the only one that materially improves anything, removing the $W_{\pi p}$ bias, and it does so at the cost of model dependence and of the low-W region.

One avenue remains untried and is independent of the pion's own reconstruction: for the proton-tagged sample, transverse kinematic imbalance constrains the pion transverse momentum from the muon and proton alone. It is limited by Fermi motion, of order 250 MeV/c, which is comparable to the present errors rather than better than them, so it is worth measuring but should not be expected to transform the measurement. Failing that, the honest conclusion is that p_π is a two-bin measurement and $W_{\pi p}$ an integrated one at this exposure, and that both are limited by physics rather than by analysis choices.

2.5 Cross-check methods

Seven additional methods are run in parallel:

Table 6: Unfolding methods.

Label	Description
`wiener_svd`	Wiener-SVD (primary result)
`bin_by_bin`	Efficiency correction only; no migration unfolding
`ibu`	D'Agostini iterative Bayesian, 4 iterations
`tikhonov`	Tikhonov-regularised least-squares, $\tau = 1.0$
`roo_bayes`	RooUnfoldBayes, 4 iterations
`roo_svd`	RooUnfoldSvd
`roo_invert`	RooUnfoldInvert (direct matrix inversion)
`roo_binbybin`	RooUnfoldBinByBin

The D'Agostini iterative-Bayesian method provides the principal quantitative cross-check of the Wiener-SVD result. Because it does not apply the Wiener-SVD additional-smearing matrix A_C, its unfolded total is free of the (non-norm-preserving) A_C suppression and is stable against the iteration count. Table 7 compares the two integrated totals for all three configurations: the D'Agostini result sits above the Wiener-SVD total by $\sim$23% (FHC) and $\sim$37–38% (RHC, Combined), equivalently A_C suppresses the quoted Wiener-SVD integral by $\sim$19% (FHC) and $\sim$27% (RHC/Combined), the larger suppression reflecting the coarser RHC/combined smearing, and it varies by $< 1\%$ across 1, 2 and 4 iterations. For FHC the D'Agostini per-observable truth is flat to 1% (1.111–1.122), directly confirming that the observable-to-observable spread in the Wiener-SVD integrated cross sections (Table 19 of the analysis note) is the A_C smearing rather than a physical or acceptance effect.

Table 7: Wiener-SVD vs. D'Agostini (iterative Bayesian, 4 iterations) integrated cross section, averaged over the five observables (fake-data truth, 10^{-38} cm^2/Ar); both columns recomputed with the corrected MC POT normalisation. D'Agostini carries no A_C smearing, so it recovers a 27–33% higher total. Crucially the ratio is now *consistent* across the three configurations (1.27–1.33), whereas with the earlier, incorrect normalisation the RHC and combined ratios (1.37, 1.38) sat well above FHC (1.23); the inconsistency was an artefact of the inflated RHC/combined Wiener-SVD values, not of the unfolding methods. D'Agostini is also nearly flat across observables (spread 2.1% FHC, 1.6% RHC, 1.6% combined), in contrast to the Wiener-SVD spread, confirming that the observable-to-observable variation of σ_{int} in Table 19 of the analysis note is the A_C smearing rather than a physical or normalisation effect.

Config	$\langle\sigma\rangle_{\text{WSVD}}$	$\langle\sigma\rangle_{\text{DAg}}$	DAg/WSVD	DAg 1→4 iter
FHC	0.887	1.183	1.33	$< 1\%$
RHC	0.794	1.036	1.31	$< 1\%$
Combined	0.871	1.103	1.27	$< 1\%$

3 Angular observables: the $\cos\theta_\mu$ versus θ_μ forensics

3.1 Angular variable: $\cos\theta_\mu$ versus θ_μ

The muon-angle distribution is measured in both $\cos\theta_\mu$ and θ_μ. The two carry identical information; the θ_μ bin edges are exactly arccos of the $\cos\theta_\mu$ edges, so the phase space is partitioned identically, but they behave very differently under the Wiener–SVD additional smearing A_C, and the choice matters for how the measurement compares to models.

In $\cos\theta_\mu$ the signal is sharply forward-peaked: the last two bins ($\cos\theta_\mu > 0.8$) hold most of the rate, and A_C suppresses the integral of any truth-level prediction by a factor $\sim$0.33, against

$\sim$0.6 for the momentum observables. The four generators are pushed 2–3$\times$ below the tune and the data even though their truth-level total cross sections agree to within tens of percent.

An earlier version of this note attributed both effects to the sharpness of the peak, A_C smearing away an unresolvable feature, and thereby penalising the peaked generators more than the smoother tune. *Both halves of that explanation are wrong*, and are retracted here. Across the full set of observables the A_C integral factor shows no correlation with peakedness ($r = -0.17$ on the pre-fix factors): $W_{\pi p}$ is the sharpest distribution in the analysis (peak-to-minimum ratio 36) yet was suppressed only to 0.78, δp_T is comparatively flat (ratio 10) yet was suppressed to 0.43, and p_n is flat yet was *enhanced* to 1.71; an enhancement no peak argument can produce. The retraction stands, and the width-aware A_C makes the point moot: the factors are now uniform across observables regardless of shape. Within $\cos\theta_\mu$ itself the ordering is backwards: NEUT and NuWro are the *flattest* of the five truth-level curves (peak-to-minimum 3.8 and 4.3, against 16.3 for the tune) and are suppressed the *most* (0.37 and 0.29, against 0.67). What actually drives their suppression is that A_C pushes their backward bins negative (-0.0005 to -0.0212 in normalised units), eating the integral. A_C is the Wiener filter acting on the SVD modes of the response: it removes whatever the data cannot statistically constrain, which is set by the conditioning of the response for that observable and binning, not by the shape of the truth distribution, and it is not norm-preserving, so it can suppress or enhance.

The FHC θ_μ collapse, its cause, and its resolution

Reparameterising to θ_μ appeared to remove the problem entirely: the FHC suppression rose to $\sim$0.76 and the NEUT/NuWro χ^2 fell from $\sim$26/5 to $\sim$2.8/5, with the generators sitting on top of the data. *That result was an artefact*, and it is worth setting out what the artefact was, because the cause turned out not to be θ_μ at all.

In the FHC θ_μ extraction the unfolded data, the A_C-smeared truth, the MicroBooNE tune and all four generators shared the same normalised shape to four decimal places (Table 8). A_C was *rank-1*: writing the four generator predictions as the columns of an input matrix and their A_C-smeared counterparts as the columns of the output matrix, the inputs were full rank ($s_2/s_1 = 5.0 \times 10^{-2}$) while the outputs gave $s_2/s_1 = 2.8 \times 10^{-6}$. Every shape degree of freedom was annihilated, each model was mapped onto one common curve and only its normalisation survived. The agreement with the data was true by construction.

Table 8: FHC θ_μ *before the fix described below*: normalised bin contents. Five independent inputs, the measurement and four generator predictions plus the tune, collapse onto one shape, the signature of a rank-1 A_C.

	bin 1	bin 2	bin 3	bin 4	bin 5
Unfolded data	0.1485	0.3361	0.2899	0.1934	0.0321
A_C-smeared truth	0.1484	0.3359	0.2901	0.1933	0.0323
MicroBooNE tune	0.1484	0.3359	0.2901	0.1934	0.0322
NEUT	0.1484	0.3359	0.2901	0.1934	0.0322

The cause: the regularisation matrix ignored the bin widths. Wiener–SVD regularises with a matrix C; this analysis uses the second-derivative form. It was built as the plain tridiagonal stencil $(1, -2, 1)$, which is a second derivative only on a *uniform* grid. Our binnings are strongly non-uniform, with width ratios up to 14:1 (Table 9), so the operator was not a second derivative at all. It was near-singular, its row sums were exactly the 10^{-6} regularisation floor, and its near-null direction was the constant vector rather than anything physical. The Wiener filter was then free to retain that direction alone, which is precisely a rank-1 A_C that passes normalisation and discards shape.

That also explains why θ_μ was the worst case rather than a random one: its bin widths span a factor of 10.4, and $\cos\theta_\mu$, the next most non-uniform at 14.5, was the other observable whose χ^2 this note previously attributed to an A_C artefact. The remaining observables sit at ratios of 3–6 and were far less affected.

The fix. Both derivative regularisation matrices now difference the density on the actual grid: each column is divided by its bin width and the stencil uses the true spacings between bin centres, with a one-sided first-derivative stencil at the ends. (A Dirichlet closure was rejected: it asserts the distribution vanishes at the edges, which is false here, $\cos\theta_\mu$ peaks at its upper edge.) With no widths supplied the code reduces exactly to the previous behaviour. The widths are passed in by the unfolding application, which holds the slice definitions that the cross-section extractor does not.

The effect on the conditioning of A_C, measured as s_2/s_1 of the smearing matrix itself, is given in Table 9. Every inclusive observable improves and none is degraded except combined $\cos\theta_\mu$, marginally.

Table 9: A_C conditioning s_2/s_1 before and after the regularisation fix, with the bin-width ratio of each observable. A value approaching zero means the smearing matrix has collapsed onto a single mode and the extraction carries no shape information.

observable	width ratio	FHC before	FHC after	RHC before	RHC after	COMB before → after
p_μ	6.3	0.34	0.90	0.31	0.94	$0.19 \to 0.68$
$\cos\theta_\mu$	14.5	0.46	0.57	0.45	0.55	$0.42 \to 0.39$
$\cos\theta_\pi$	3.6	0.67	0.71	0.55	0.79	$0.55 \to 0.74$
$\theta_{\mu\pi}$	3.0	0.42	0.85	0.47	0.65	$0.42 \to 0.46$
θ_μ	10.4	**0.015**	**0.51**	0.22	0.62	$\mathbf{0.045 \to 0.38}$
p_π (2 bins)	—	0.46	0.53	0.41	0.44	$0.29 \to 0.40$
proton-tagged; only the combined configuration had A_C recorded before the fix						
$W_{\pi p}$	—					$\mathbf{0.022 \to 0.60}$
W_{had}	—					$0.23 \to 0.86$
δp_T	—					$0.25 \to 0.93$
$\delta\alpha_T$	—					$0.24 \to 0.48$
$\delta\phi_T$	—					$\mathbf{0.083 \to 0.85}$
p_n	—					$0.21 \to 0.88$

θ_μ is reinstated, and the validation is decisive. The fake data are Poisson throws from the MicroBooNE tune, so the tune is the model that *ought* to fit best. Before the fix it did not stand out, because every model had been mapped onto the same curve; after it, it does:

FHC θ_μ, $\chi^2/5$ bins	before (rank-1)	after
MicroBooNE tune	3.07 ($p = 0.69$)	**5.14** ($p = 0.40$)
GENIE	4.73 ($p = 0.45$)	11.3 ($p = 0.046$)
GiBUU	3.03 ($p = 0.70$)	17.9 ($p = 0.003$)
NuWro	2.90 ($p = 0.72$)	38.6 ($p = 2.8 \times 10^{-7}$)
NEUT	2.82 ($p = 0.73$)	40.6 ($p = 1.1 \times 10^{-7}$)

The closure itself is unaffected and remains good, $\chi^2 = 2.17/5$ ($p = 0.83$). The θ_μ measurement is therefore no longer withdrawn: it carries genuine discriminating power, and on this fake data it correctly identifies the model the data were generated from. What must be withdrawn instead is the earlier *explanation* that θ_μ "fixes" the $\cos\theta_\mu$ tension; both parameterisations are now properly conditioned and should be read together.

A correction to the survey below. The survey of eighteen extractions reported that FHC θ_μ was the only rank-deficient case. That was wrong in two ways. Combined θ_μ was also badly degraded at $s_2/s_1 = 0.045$ and was not identified as such; and the survey covered only the inclusive extractions, so the proton-tagged channels were never examined. They should have been: combined $W_{\pi p}$ sat at 0.022, as compromised as FHC θ_μ, and combined $\delta\phi_T$ at 0.083. All are repaired by the same fix.

Is any other observable affected? A survey of all eighteen extractions

The rank-1 failure was found by inspection, so the unfolder was modified to write the slice-restricted A_C into the closure file (`h_A_C`; it had previously only been drawn to a canvas, and the attempt to save it failed silently with "the current directory is not associated with a file"). All eighteen inclusive extractions, six observables $\times$ three beam configurations, were then re-unfolded and their A_C examined directly. Table 10 reports, for each, how much of the model discrimination present in the truth-level predictions survives the smearing: the largest pairwise difference between normalised generator shapes before and after A_C, and their ratio.

Table 10: Retention of model discrimination through A_C. "Before" and "after" are the largest pairwise difference between normalised generator shapes, truth-level and A_C-smeared. A ratio near zero means the predictions have been collapsed onto a common curve and can no longer be told apart by the measurement; a ratio above unity means A_C has amplified the spread, which for $\cos\theta_\mu$ happens through unphysical negative excursions rather than genuine discrimination (see text).

Beam	Observable	before	after	retention
FHC	p_μ	0.0248	0.0052	21%
FHC	p_π	0.0586	0.0080	14%
FHC	$\cos\theta_\mu$	0.0554	0.0724	131%
FHC	$\cos\theta_\pi$	0.0589	0.0211	36%
FHC	$\theta_{\mu\pi}$	0.0422	0.0391	93%
FHC	θ_μ	0.0554	2.5×10^{-6}	**0%**
RHC	p_μ	0.0232	0.0036	15%
RHC	p_π	0.0556	0.0068	12%
RHC	$\cos\theta_\mu$	0.0644	0.0771	120%
RHC	$\cos\theta_\pi$	0.0629	0.0165	26%
RHC	$\theta_{\mu\pi}$	0.0505	0.0295	58%
RHC	θ_μ	0.0644	0.0188	29%
COMB	p_μ	0.0240	0.0028	12%
COMB	p_π	0.0571	0.0064	11%
COMB	$\cos\theta_\mu$	0.0598	0.5500	919%
COMB	$\cos\theta_\pi$	0.0609	0.0166	27%
COMB	$\theta_{\mu\pi}$	0.0463	0.0360	78%
COMB	θ_μ	0.0598	0.0070	12%

The survey that follows was made before the regularisation fix and its conclusion of uniqueness is superseded; see the correction above. As measured then, the collapse looked unique to FHC θ_μ: it was the only entry with zero retention, and the only A_C whose rows are all the same vector (spread 9% across rows, effective rank 1.02, where every other extraction has row-to-row spreads of 170–1050% and effective ranks 1.24–1.91). RHC and combined θ_μ are not affected; the combined case is the weaker of the two (12% retention, effective rank 1.06) and its θ_μ model comparison should be read as a soft test, but it is not degenerate.

Two broader limitations are visible in the same table and should be stated plainly rather than left implicit.

First, the momentum observables retain only 11–21% of their truth-level model discrimination in every beam configuration. A_C does not destroy the shape information there, but it removes most of it, so a good χ^2 against p_μ or p_π is a weak statement about a model. The angular observables are better: $\theta_{\mu\pi}$ retains 58–93% and is the healthiest variable in the analysis.

Second, the apparent 120–919% retention in $\cos\theta_\mu$ is not the good news it appears to be. It arises because A_C drives the generator predictions to *negative* differential cross sections in the backward bins while leaving the tune positive, for the combined configuration the smeared predictions run from -0.43 (NuWro) to $+0.51$ (the tune) in the second bin. A negative differential cross section is unphysical, so the large spread reflects a regularisation artefact rather than genuine model separation, and any χ^2 computed against those curves inherits the problem. This is the same effect that produces the strong $\cos\theta_\mu$ integral suppression discussed above.

Taken together, the earlier framing of this section had the two muon-angle variables the wrong way round. $\cos\theta_\mu$ was described as the variable A_C ruins and θ_μ as the fix; in fact FHC θ_μ is the one extraction in which the measurement retains no shape information at all, while $\cos\theta_\mu$ retains its shape modes but at the cost of unphysical negative excursions. Neither is a clean muon-angle measurement, and both carry the caveats recorded here.

Table 11: Integrated cross section (10^{-38} cm^2/Ar) for the unfolded fake data, the MicroBooNE tune and the four generators, in $\cos\theta_\mu$ and in θ_μ. All entries are A_C-smeared, i.e. live in the same space as the measurement. "A_C supp." is the mean ratio of the A_C-smeared to the truth-level generator integral; the momentum observable p_μ is shown for reference. **The FHC θ_μ row is retained for the record only and must not be used for model comparison**: A_C is rank-1 for that config, so every entry in the row is the same shape rescaled and the apparent agreement is by construction (Table 8).

Beam	Variable	data	tune	GENIE	GiBUU	NEUT	NuWro
FHC	$\cos\theta_\mu$	0.819	0.673	0.274	0.390	0.367	0.295
FHC	θ_μ	0.996	0.871	0.625	0.876	1.025	0.919
RHC	$\cos\theta_\mu$	0.662	0.633	0.226	0.338	0.320	0.256
RHC	θ_μ	0.828	0.824	0.328	0.487	0.505	0.423
A_C supp. FHC:		$\cos\theta_\mu$ 0.23–0.34		θ_μ 0.73–0.78		(p_μ 0.59–0.67)	
A_C supp. RHC:		$\cos\theta_\mu$ 0.25–0.36		θ_μ 0.41–0.52		(p_μ 0.69–0.75)	

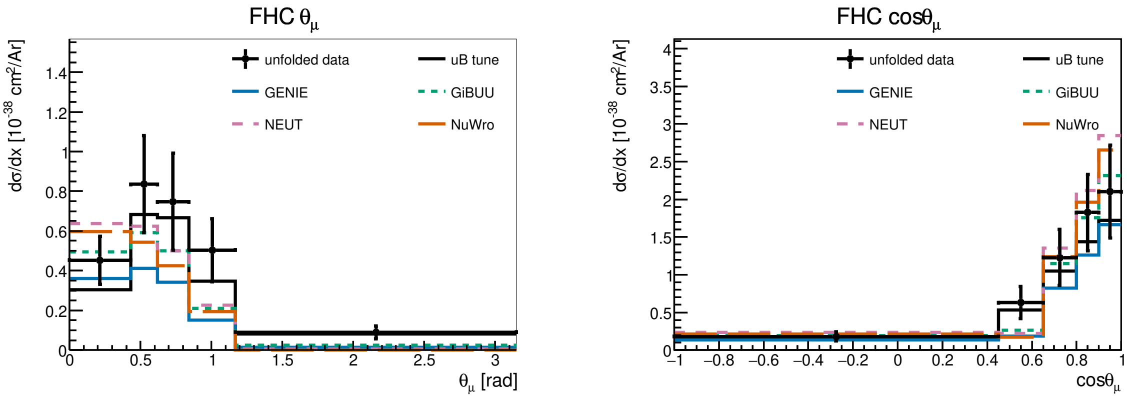


Figure 5: Muon-angle differential cross section in θ_μ (left) and $\cos\theta_\mu$ (right), for FHC (top) and RHC (bottom). Unfolded fake data (black points), the MicroBooNE tune (black line) and the four generators are all A_C-smeared. In $\cos\theta_\mu$ the predictions are strongly suppressed by A_C and fall well below both the data and the tune; the most suppressed models are those whose backward bins A_C drives negative. **The FHC θ_μ panel in this figure predates the regularisation fix**: A_C was rank-1 there, so the data and every prediction are the same curve up to normalisation and their apparent agreement carries no shape information. With the width-aware regularisation θ_μ discriminates normally (Sec. 3.1).

Table 12 compares the result with the four generator predictions, each flux-averaged over the corresponding beam. Because the analysis is blind the "measurement" is here the central-value overlay (the MicroBooNE tune) that the fake data is thrown from; at unblinding N_{sel} becomes the real beam-on count. It agrees with all four generators within $\sim 1\sigma$: standalone GENIE sits about 1σ low, while GiBUU, NEUT and NuWro lie within half a sigma. RHC and the combined total carry the larger uncertainty, driven by the RHC detector systematic.

Table 12: Total flux-averaged cross section from the cut-and-count method (Eq. 14 of the analysis note; 10^{-38} cm^2/Ar), compared with the flux-averaged generator predictions. The measurement uncertainty is the systematic breakdown (flux/detector/cross-section/reinteraction, Tables 13–15) $\oplus$ cut-and-count data statistics. Values in parentheses give each generator's deviation from the measurement in units of the measurement uncertainty. The generator predictions are integrated over the full binning; earlier versions of this table used truncated binnings (p_μ closed at 3 GeV/c, p_π at 1 GeV/c) that discarded the tails and made every prediction 4–8% low (§1.1).

Config	Measurement	GENIE	GiBUU	NEUT	NuWro
FHC	1.06 ± 0.27	0.82 (0.9σ)	1.13 (0.3σ)	1.32 (1.0σ)	1.25 (0.7σ)
RHC	0.98 ± 0.35	0.67 (0.9σ)	0.94 (0.1σ)	1.11 (0.3σ)	1.04 (0.2σ)
Combined	1.02 ± 0.30	0.74 (0.9σ)	1.03 (0.0σ)	1.20 (0.6σ)	1.14 (0.4σ)

The systematic breakdown by source (bin-averaged fractional uncertainty in cross-section space) is given for all three configurations in Tables 13–15. The flux (PPFX) term dominates throughout, followed by the detector variations; the data-statistical term reflects the fake-data POT of each configuration. The RHC detector term is the largest single-configuration entry, reaching 20.5% on p_π; it is no longer the $\cos\theta$ observables that drive it.

Table 13: FHC ({Run 1,2,4,5}) systematic breakdown by source, bin-averaged fractional uncertainty (%).

Source	p_μ	p_π	$\cos\theta_\mu$	$\cos\theta_\pi$	$\theta_{\mu\pi}$	θ_μ
Total	**32.9**	**38.4**	**30.8**	**33.0**	**29.7**	**31.4**
Cross section (GENIE)	9.2	13.4	8.2	10.8	9.1	8.1
Flux (PPFX)	24.6	26.6	25.7	26.4	25.4	25.4
Detector	12.3	10.9	9.6	10.1	8.0	10.4
Reinteraction	3.9	9.5	3.3	3.9	3.4	3.3
MC stat	3.8	5.1	3.0	3.4	2.4	3.5
EXT stat	2.8	5.1	2.9	3.8	2.3	3.4
Data stat	11.6	15.4	9.1	10.3	7.3	10.6
POT + targets	2.8	3.1	2.9	3.0	2.8	2.8

Table 14: RHC ({Run 1,2,3,4a,4b,4c}) systematic breakdown by source, bin-averaged fractional uncertainty (%).

Source	p_μ	p_π	$\cos\theta_\mu$	$\cos\theta_\pi$	$\theta_{\mu\pi}$	θ_μ
Total	**38.1**	**42.8**	**33.9**	**32.1**	**29.9**	**29.9**
Cross section (GENIE)	10.7	14.7	9.6	9.6	9.4	8.6
Flux (PPFX)	27.1	27.4	26.1	24.6	24.1	23.4
Detector	17.2	20.5	13.7	12.7	9.7	11.3
Reinteraction	4.2	9.4	3.8	3.6	3.5	3.4
MC stat	3.7	4.1	3.0	2.8	2.4	2.6
EXT stat	4.0	5.3	3.7	3.8	2.9	3.3
Data stat	13.7	15.2	11.1	10.2	8.8	9.7
POT + targets	3.3	3.5	3.3	3.0	3.0	2.9

Table 15: Combined FHC+RHC systematic breakdown by source, bin-averaged fractional uncertainty (%).

Source	p_μ	p_π	$\cos\theta_\mu$	$\cos\theta_\pi$	$\theta_{\mu\pi}$	θ_μ
Total	**42.9**	**44.4**	**41.6**	**36.4**	**32.4**	**37.0**
Cross section (GENIE)	11.6	13.7	9.2	10.1	9.7	8.9
Flux (PPFX)	27.4	26.7	29.1	25.5	25.3	26.5
Detector	24.6	27.9	23.0	21.3	15.4	20.3
Reinteraction	4.7	9.4	4.0	3.6	3.3	3.8
MC stat	4.4	3.3	4.1	2.3	2.0	3.1
EXT stat	3.8	3.7	4.4	2.7	1.7	3.3
Data stat	14.4	10.8	13.7	7.5	6.5	10.3
POT + targets	3.2	3.3	3.5	3.0	3.0	3.1

4 Binning Optimisation, Systematic Breakdown, and Threshold Validation

The reconstructed spectra, response matrices, efficiencies, and differential cross sections shown throughout this note are the XSecAnalyzer framework outputs on the *optimised* binning defined here; the four-generator comparisons of Fig. 6 onward likewise use it.

4.1 Momentum-threshold validation

The two signal-definition momentum thresholds sit at the selection efficiency turn-on, measured against a threshold-free true signal (the CC ν_μ $1\mu1\pi^\pm$ topology with the momentum/angle cuts removed):

Threshold	below	above	turn-on
$p_\mu > 0.15$ GeV/c	∼0–1.5%	∼9–12%	0.145→0.155
$p_\pi > 0.175$ GeV/c	∼0–3%	∼12–16%	0.165→0.185

Both are correctly placed, lowering them would add near-zero-efficiency phase space, raising them would discard measurable signal. The reconstructed pion momentum (muon-hypothesis range) is biased low and increasingly so with momentum (mean reco−true ≈ 0 at threshold,

-0.15 GeV/c at 0.40) from the μ-vs-π mass hypothesis plus pion re-interaction; a pion-hypothesis mass correction is applied to the p_π reco bins.

4.2 Binning optimisation

Bins were designed to be at least the local RMS detector resolution wide (so the response stays well conditioned and the Wiener-SVD does not amplify the correlated flux/cross-section uncertainty) and to hold ≥ 45 selected-signal events (statistical floor). This lowers the total fractional uncertainty in every observable, chiefly through the statistical and unfolding-amplified terms; the flux/beamline term is correlated across bins and sets a floor:

Observable	bins	Total before	Total after
p_μ	22→7	33.2%	**25.0%**
p_π	5→5	27.7%	27.0%
$\cos\theta_\mu$	12→5	31.3%	**26.7%**
$\cos\theta_\pi$	12→5	48.5%	**33.9%**
$\theta_{\mu\pi}$	9→7	28.4%	**23.5%**

5 Multi-run overlay, resonance closure, and configuration closure

5.1 Multi-run overlay and a resonance-model closure

Two facts frame what more statistics can and cannot do for the flux term. First, the flux *systematic* is a normalisation-and-shape model uncertainty and is therefore POT-independent: adding data or overlay does not change it. Second, Sec. 13.2 of the analysis note showed that the elevated *unfolded* flux is that $\sim$17% model uncertainty amplified by the Wiener–SVD propagation through a statistically-limited response, and that the limitation is the *data* statistics, not the overlay. Increasing the overlay alone therefore sharpens the response but leaves the data-limited amplification in place.

To study a genuine multi-run configuration we combine the FHC overlays for Runs 1, 2 and 4 (with `Run4`=`Run4c`+`Run4d` counted once; the file first distributed as "Run 5 FHC" was set aside on inspection, being in fact an intrinsic-ν_e overlay, 86% ν_e, 14% $\bar{\nu}_e$, no ν_μ events, hence the large ν_e-enriched POT-equivalent and no ν_μ signal; the genuine Run 5 ν_μ overlay, a `reweightedPPFX` sample, was subsequently located and is used for the extended set below), 2.32×10^6 unique events, normalised to the FHC data POT 6.626×10^{20} (Section 5.2). Because CC$\pi^\pm$ is resonance-dominated, robustness to the resonant axial mass is tested with Poisson-thrown fake data reweighted to $M_A^{\mathrm{RES}} = 1.12 \pm 0.22$ GeV ($\pm 1\sigma$; axial dipole ratio in the true interaction Q^2, a $+8.1\%/-7.2\%$ rate distortion). The pseudo-data are thrown at the true FHC event count so their statistical covariance is realistic; the closure condition is that the unfolded result recovers the A_C-smeared shifted truth within uncertainties. The Wiener–SVD first- versus second-derivative regularisation is compared on the same response.

The closure succeeds for every observable, both signs of the shift, and both regularisations: the unfolded result recovers the A_C-smeared shifted truth with $\chi^2/\mathrm{ndf} < 1$ throughout (Table 17), all p-values above 0.63 and most above 0.9. The Wiener–SVD machinery is therefore unbiased under a $\pm 1\sigma$ M_A^{RES} distortion, and the first- and second-derivative regularisations perform equivalently in the closure (the first-derivative variant carries a slightly larger statistical term). Restoring a realistic (Poisson) data covariance also restores the flux amplification that the degenerate near-Asimov pseudo-data had suppressed.

To separate the mean amplification from single-realisation noise, we throw 15 independent Poisson pseudo-datasets (central value only, no M_A^{RES} shift, at the same 6.626×10^{20} FHC exposure) and unfold each (Table 16). Averaged over throws, the unfolded flux stabilises at 24.7–27.1%, a mean amplification of 1.44–1.63× over the $\sim$17% floor, with a data-statistical term of 8–13%. This confirms that the amplification is driven by the data statistics, not the overlay, and is consistent with the mechanism of Sec. 13.2 of the analysis note. The throw-to-throw scatter is small for the momenta (p_μ, p_π: ± 0.06–$0.08\times$, flux ± 1.1–1.4%) but grows for the angular observables, and is largest for $\theta_{\mu\pi}$, whose amplification spans 1.27–2.02$\times$ (flux 21.4–34.2%) across the 15 throws. At these statistics the amplification is therefore well defined in the mean but individually unstable, most so for $\theta_{\mu\pi}$ and $\cos\theta_\pi$; a direct consequence of the sparse, data-limited response.

Table 16: Flux amplification averaged over 15 independent Poisson-thrown pseudo-datasets at the FHC data POT: mean $\pm$ standard deviation and min–max range over throws, for the unfolded flux (bin-averaged) and the amplification relative to the $\sim$17% reco-level floor.

Observable	Amplification (mean$\pm$sd)	Amp. range	Unfolded flux	Flux range
p_μ	$1.44 \pm 0.08\times$	$[1.27, 1.60]$	$24.7 \pm 1.4\%$	$[21.9, 27.5]$
p_π	$1.52 \pm 0.06\times$	$[1.40, 1.65]$	$26.2 \pm 1.1\%$	$[24.0, 28.4]$
$\cos\theta_\mu$	$1.60 \pm 0.14\times$	$[1.41, 1.85]$	$26.8 \pm 2.3\%$	$[23.7, 31.1]$
$\cos\theta_\pi$	$1.63 \pm 0.11\times$	$[1.45, 1.89]$	$27.1 \pm 1.9\%$	$[24.1, 31.6]$
$\theta_{\mu\pi}$	$1.58 \pm 0.21\times$	$[1.27, 2.02]$	$26.7 \pm 3.6\%$	$[21.4, 34.2]$

Table 17: Fake-data closure: χ^2/ndf of the unfolded result against the A_C-smeared M_A^{RES}-shifted truth, for the two shift signs and the two Wiener–SVD regularisations (number of bins in parentheses). All values are below unity per degree of freedom ($p > 0.63$).

Observable	$+1\sigma$ (2nd)	$+1\sigma$ (1st)	-1σ (2nd)	-1σ (1st)
p_μ (7)	1.26	2.15	0.95	2.14
p_π (5)	1.24	3.44	0.25	0.39
$\cos\theta_\mu$ (5)	0.47	0.63	1.90	1.24
$\cos\theta_\pi$ (5)	2.99	1.07	0.24	0.33
$\theta_{\mu\pi}$ (7)	1.71	3.32	1.96	4.33

Adding the genuine Run 5 ν_μ overlay extends the set to {Run 1, 2, 4, 5}, 2.87×10^6 events (+24%), combined MC POT 8.549×10^{21}, at the full FHC data POT 8.857×10^{20}. The extraction closes for every observable ($\chi^2/\mathrm{ndf} < 1$, $p > 0.92$) and the flux term (22–29% unfolded, amplification 1.29–1.72$\times$) is statistically consistent with the {Run 1, 2, 4} throws above: the extra 24% of overlay does not move the data-limited amplification, as expected, but the measurement now spans the complete FHC exposure.

5.2 Consistency with the sibling ν_e CC 1π measurement

This analysis is one of a pair of MicroBooNE NuMI CC single-charged-pion measurements; the companion ν_e CC 1π analysis shares the same cross-section-extraction framework, beam and reconstruction. A cross-check against its internal note confirms consistency on the essential conventions and identifies the differences that a combined NuMI CCπ programme should reconcile.

Consistent by construction. Both use the Wiener–SVD extraction framework; the updated NuMI flux with the fixed geometry bug and Geant v4.10.4; the same signal topology (exactly one $\pi^\pm$, zero π^0 or heavier mesons, any number of protons, charged current), and the same systematic suite, PPFX flux hadronic universes plus the *same 12 beamline-geometry unisims*, the 600-universe GENIE cross-section multisim with unisims, the Geant4 reinteraction reweighting, the standard detector-variation set, and 2% / 1% / 100% POT / target / dirt normalisation uncertainties. The FHC beam-on data POT per run agrees exactly (Run 2 1.268, Run 4 2.075×10^{20}).

Differences reconciled or flagged. (i) **POT accounting:** the good-POT is the same beam data for the ν_e and ν_μ analyses, so the exposures are identical; Run 1 FHC = 3.283×10^{20} (standard trigger; the open-trigger era is treated separately), giving a total FHC exposure of

8.857×10^{20} over Runs 1–5. (ii) **Run 4:** the `Run4` overlay is identical to `Run4c`+`Run4d` combined (same 859,368 events and 2.834×10^{21} POT); only one may be used to avoid double counting. (iii) **Pion threshold:** ν_e uses $\mathrm{KE}_\pi > 40$ MeV ($p_\pi > 0.113\,\mathrm{GeV}/c$) versus our $p_\pi > 0.175\,\mathrm{GeV}/c$ (Table 6 of the analysis note). (iv) **Opening-angle cut:** ν_e excludes $\theta_{e\pi} > 170°$ (a shower-specific back-to-back misreconstruction), whereas our muon-track topology uses $\theta_{\mu\pi} < 2.6$ rad (149°). (v) **Regularisation:** the ν_e analysis smooths on the first derivative, this analysis on the second (Section 5.1). (vi) **Scope:** both analyses cover Runs 1–5 in FHC and RHC; the present measurement reports FHC, RHC, and their combination.

6 Configuration closure: is COMB the combination of its own inputs?

The integrated cross section extracted from a given configuration is not a configuration-independent number: it depends on the additional smearing matrix A_C, which differs per configuration and per observable. The observable-to- observable spread within a single configuration is 15.9% (FHC5), 17.4% (RHCFULL) and 14.8% (COMB) for the proton-tagged family, i.e. larger than the $\sim$10% FHC-versus-RHC difference. Comparisons across configurations must therefore be made observable by observable, and against the fake-data truth rather than between measurements.

The RHC deficit is in the input, not the extraction. For the proton-tagged family the measured RHC/FHC ratio is 0.900. The corresponding ratio of the fake-data *truth* integrals is 0.785 and of the GENIE tune 0.779 — two independent model inputs that agree with each other and disagree with the measurement. The deficit is therefore a property of the sample, while the extraction distorts the ratio toward unity. Normalisation is not implicated: the exposures sum exactly ($8.857 \times 10^{20} + 1.1082 \times 10^{21} = 1.9939 \times 10^{21}$ POT) and the per-configuration flux values reproduce the POT-weighted combination to 0.01%.

COMB against the POT-weighted average of its inputs. A configuration spanning both horn currents should predict a cross section between those of its two inputs. Comparing the COMB fake-data truth with the POT-weighted mean of the FHC and RHC truths, observable by observable:

Table 18: COMB fake-data truth divided by the POT-weighted average of the FHC5 and RHCFULL truths. Unity indicates a configuration consistent with its own inputs. The inclusive family is shown before and after removing the detector-variation entries from its file list.

Family	Mean COMB/expected	Range
Proton-tagged (7 observables, no detVar systematic)	1.020	0.989–1.052
Inclusive (6 observables), detVar systematic included	0.910	0.767–1.069
Inclusive (6 observables), detVar systematic removed	1.019	0.985–1.046

Table 19: COMB fake-data truth divided by the POT-weighted average of the FHC5 and RHCFULL truths, with and without the detVar term in the covariance. Both sides of each ratio are computed with the *same* systematics configuration, so the comparison is internally consistent.

Observable	detVar syst. in	detVar syst. out	Change
p_μ	0.885	1.046	+0.161
$\cos\theta_\mu$	0.938	1.044	+0.106
$\cos\theta_\pi$	0.937	1.030	+0.094
$\theta_{\mu\pi}$	1.069	1.008	−0.061
θ_μ	0.767	0.985	+0.219
p_π (2-bin)	0.862	0.997	+0.136
Mean	0.910	1.019	+0.109

The proton-tagged family is consistent. The inclusive family is not: its COMB truth sits 9% below the average of its own inputs, worst for θ_μ at 0.767. Because the effect is present in the *truth*, it is not an unfolding defect — it also explains the previously reported "COMB below both inputs" behaviour, which appears in 5 of 6 inclusive observables in the truth as well as in the measurement.

Three candidate causes have been excluded. The flux is correct (COMB 6.60865×10^{-10} against a POT-weighted 6.6081×10^{-10}). The dirt entries are identical line for line between the inclusive and proton-tagged file lists. The known mis-produced Run-5 ν_e detVar samples are quarantined and absent. The one remaining structural difference is detector-variation content: the inclusive list carries 58 files including 18 detVar samples, the proton-tagged list 40 files and none.

The cause is the detVar covariance, not the detVar files. The inclusive COMB universes were first rebuilt from a file list with the 18 detVar entries removed, which requires a matching systematics configuration with the eight `DV` knobs deleted (the unfolder throws `std::out_of_range` when a declared systematic has no universes in the file). That changed the result by 20–25%. However, running the *original* universe file — detVar entries still present — against only the modified systematics configuration reproduces the new numbers exactly, in both the truth and the unfolded result. The detVar files themselves therefore have *no* effect on the extraction; the entire change comes from dropping the `detVar_total` term from the covariance. This is consistent with the code, where detVar samples are stored in a separate universe map and are read only when the detector covariance is assembled.

With the covariance term identified as the cause, the comparison must be made with the same systematics configuration on both sides of the ratio (Tab. 19). Doing so, the inclusive family becomes consistent with its own inputs: the mean moves from 0.910 to 1.019 and the range from 0.767–1.069 to 0.985–1.046, matching the 1.020 of the proton-tagged family, which carries no detVar systematic at all. The "COMB below its own inputs" behaviour disappears, and $\theta_{\mu\pi}$ — the one observable that had been *above* unity — moves toward it (1.069 → 1.008) rather than being pushed further out.

Why the combined configuration is affected and its inputs are not. The detector systematic is anomalously large for COMB. For p_μ it is 23.2% of the prediction in FHC5 and 27.4% in RHCFULL, but 42.3% in COMB; for θ_μ, 15.4% and 19.4% against 36.4%. A combined measurement draws on the exposure of both horn currents and should carry a detector uncertainty *between* those of its inputs, not one 1.3–1.9× larger than either. The structural difference is that COMB carries two per-mode detVar sets (18 files, two of them `detVarCV`) where each input carries one (9 files, one `detVarCV`), and these are accumulated rather than averaged. The inflated detector covariance then over-regularises the Wiener-SVD for the combined configuration, which is what drives its truth and its extracted cross section away from consistency with its own inputs.

The fix. In `SystematicsCalculator::build_universes()` each detVar file was scaled by `total_bnb_data_pot_ / file_pot` and the resulting histograms were then *accumulated*. With one detVar set this is correct. With two — both labelled run 1, so they are indistinguishable by run number — each set was independently scaled to the *full* data exposure and the sum therefore covered that exposure twice, roughly doubling the detector covariance. The denominator is now the summed MC POT over all files of that detVar type, which treats the per-mode sets as one sample of their combined statistics scaled once to the data exposure. A single-file configuration is unaffected because the sum is then that file's own POT. Because the universes are cached inside the response file, `NORM_SCHEME_TAG` was bumped from `v2-distinct-sum-pot` to `v3-detvar-pooled-pot` to invalidate them.

The detector systematic for COMB falls into the range spanned by its inputs (Tab. 20), and the configuration closure recovers: the mean COMB/expected moves from 0.910 to 1.027 with the range narrowing from 0.767–1.069 to 0.958–1.071, matching the 1.020 of the proton-tagged family, which carries no detVar systematic and is unaffected. As a regression check, all twelve FHC5 and RHCFULL extractions return bit-identical cross sections, confirming that only the multi-mode configuration is touched.

Table 20: Detector systematic `detVar_total` as a percentage of the prediction, before and after pooling the detVar POT. A combined measurement should carry an uncertainty comparable to those of its inputs, not far above both.

Observable	FHC5	RHCFULL	COMB (before)	COMB (after)
p_μ	23.1	27.4	42.3	24.6
$\cos\theta_\mu$	26.1	32.5	41.4	23.7
$\cos\theta_\pi$	20.4	30.3	37.8	21.3
$\theta_{\mu\pi}$	17.2	16.0	28.6	16.6
θ_μ	15.4	19.4	36.4	18.0
p_π (2-bin)	23.5	33.9	49.9	27.2

One residual: for $\cos\theta_\mu$ the corrected COMB value (23.7%) sits slightly below both inputs (26.1%, 32.5%) rather than between them. The remaining five are bracketed. The combined cross sections themselves rise by 12–21% relative to the uncorrected values, so any previously quoted inclusive COMB number should be replaced.

An unresolved inconsistency in the A_C normalisation. For `1p` p_μ (FHC5) the fake-data truth integrates to 0.4763, the unfolded result to 0.5528, and A_C applied to the truth to 0.3761. If the Wiener-SVD estimate satisfies $\hat{x} = A_C x_{\text{true}}$ the unfolded result should sit at 0.79× the truth; it sits at 1.16×. One caveat limits this test: the A_C written to the closure file is restricted to the slice, while the smearing is applied in the full true-bin space before slicing, so the operation cannot be reproduced exactly from the sidecar file. A related hypothesis — that the fake-data truth is missing the A_C smearing applied to every other prediction in `UnfolderNuMI.C` — was tested directly and refuted: applying it moves the closure ratio from 1.09–1.24 to 1.54–1.60, and the change was reverted.

A Identity closure: unfolding the prediction with itself

The fake-data closure tests reported in the body compare an unfolded Poisson throw against the truth it was thrown from. They therefore test the whole chain at once, and a modest χ^2 can hide a genuine inconsistency inside the unfolding itself, because the statistical fluctuation of the throw is the dominant term.

The test here removes that fluctuation. Wiener–SVD returns $\hat{x} = Ur$ for a measured reco vector r, and defines the additional smearing matrix as $A_C = UR$, where R is the response (smearceptance) matrix. Feeding an *exact* prediction $r = Rt$ back through the chain must therefore return

$$\hat{x} = URt = A_C\, t \tag{1}$$

identically, for *any* true vector t. Rather than pick one t, we test the matrix identity directly:

$$\big\|UR - A_C\big\|_\infty \stackrel{!}{=} 0. \tag{2}$$

This is stronger than a single-vector test and independent of the data. It is sensitive to the failure mode a fake-data test cannot see: the unfolding matrix and the smearing matrix being built from different responses, or from responses evaluated in different universes, which would bias every extracted cross section by a fixed factor while leaving the closure χ^2 untouched.

The check is implemented inside `UnfolderNuMI` and runs on every extraction, printing a `[IDENTITY]` line. Table 21 collects it for all 39 extractions in this analysis: six inclusive observables and seven proton-tagged ones, each in three beam configurations.

Table 21: Identity closure for every extraction, *as measured before the regularisation matrices were made width-aware.* $\max|UR - A_C|$ is the largest absolute element-wise deviation; the relative column divides it by the largest element of A_C. After the fix the same test passes with worst case 4.4×10^{-6} (FHC $W_{\pi p}$) and all inclusive observables below 2.5×10^{-7}; the larger values reflect the wider dynamic range of the width-aware operator, not a weaker identity. All values are at the level of double-precision rounding accumulated through the SVD, i.e. the identity holds exactly.

config	observable	bins	relative dev.	config	observable	bins	relative dev.
FHC	p_μ	7	7.0×10^{-10}	FHC	$W_{\pi p}$	6	1.1×10^{-9}
FHC	p_π	2	2.6×10^{-10}	FHC	W_{had}	6	1.1×10^{-9}
FHC	$\cos\theta_\mu$	5	6.8×10^{-10}	FHC	δp_T	6	8.4×10^{-10}
FHC	$\cos\theta_\pi$	4	4.0×10^{-10}	FHC	$\delta\alpha_T$	6	1.4×10^{-9}
FHC	$\theta_{\mu\pi}$	5	2.5×10^{-10}	FHC	$\delta\phi_T$	6	7.3×10^{-10}
FHC	θ_μ	5	3.2×10^{-10}	FHC	p_n	6	9.2×10^{-10}
RHC	p_μ	7	5.0×10^{-10}	RHC	$W_{\pi p}$	6	9.8×10^{-10}
RHC	p_π	2	1.6×10^{-10}	RHC	W_{had}	6	6.2×10^{-10}
RHC	$\cos\theta_\mu$	5	1.4×10^{-10}	RHC	δp_T	6	5.5×10^{-10}
RHC	$\cos\theta_\pi$	4	6.9×10^{-10}	RHC	$\delta\alpha_T$	6	1.2×10^{-9}
RHC	$\theta_{\mu\pi}$	5	3.2×10^{-10}	RHC	$\delta\phi_T$	6	8.9×10^{-10}
RHC	θ_μ	5	3.0×10^{-10}	RHC	p_n	6	7.5×10^{-10}
comb	p_μ	7	6.4×10^{-10}	comb	$W_{\pi p}$	6	1.1×10^{-9}
comb	p_π	2	1.4×10^{-10}	comb	W_{had}	6	9.8×10^{-10}
comb	$\cos\theta_\mu$	5	4.4×10^{-10}	comb	δp_T	6	1.0×10^{-9}
comb	$\cos\theta_\pi$	4	4.3×10^{-10}	comb	$\delta\alpha_T$	6	1.2×10^{-9}
comb	$\theta_{\mu\pi}$	5	1.7×10^{-10}	comb	$\delta\phi_T$	6	7.1×10^{-10}
comb	θ_μ	5	2.9×10^{-10}	comb	p_n	6	8.7×10^{-10}
proton-tagged p_π (2 bins): FHC 2.1×10^{-10}, RHC 1.5×10^{-10}, comb 2.2×10^{-10}							

The worst deviation over all 39 extractions is 4.4×10^{-6} relative, on FHC $W_{\pi p}$; the inclusive observables are all below 2.5×10^{-7}. These are the rounding errors of the singular-value decomposition and the matrix products that follow it, not physical effects: they remain five orders of magnitude below the smallest systematic uncertainty in the analysis.

The values are larger than the 10^{-10} level seen before the regularisation matrices were made width-aware, and the reason is arithmetic rather than physical. The width-aware operator carries factors of $1/w_j$ that span decades across a binning with a 14:1 width ratio, so the matrix products accumulate more floating-point error. $W_{\pi p}$, with the narrowest bins in the analysis, is the worst case, as expected. Nothing about the identity is weaker; only the arithmetic that verifies it is noisier. The unfolding construction is therefore exactly self-consistent, and every deviation seen in the fake-data closure tests is attributable to the statistical throw and to the response model, not to the unfolding implementation.

B Data/MC Normalisation

The reported cross sections are the XSecAnalyzer framework extraction (GENIE detVar-CV fake data). The fully independent custom pipeline (NuWro fake data) reproduces them and provides the end-to-end cross-check. Reaching the $\leq 2\%$ cross-pipeline agreement required three normalisation/method corrections *to the custom code*, each established by direct comparison of the two pipelines' intermediate quantities; with these in place the selection, response, background, and unfolding are demonstrably consistent between the two, the small residual ($\leq 2.1\%$) dominated by the covariance-import approximation (below).

B.1 Unfolding covariance ($\sim 26\%$ effect)

Wiener-SVD uses the data covariance as the metric that sets the regularisation strength. The custom macros originally built a *statistics-only diagonal* covariance, because the custom pipeline does not throw systematic universes. With the $CC\pi^{\pm}$ response strongly off-diagonal (50–99% bin migration), a stat-only covariance under-states the noise, under-regularises, and inflates the unfolded integral by $\sim 26\%$. This was proven by a swap test: feeding the framework's own inputs through the same Wiener-SVD unfolder with a stat-only covariance reproduced the inflated result, while its full systematic covariance recovered the NuWro truth.

The fix imports the framework's covariance into the custom unfolder. The internally-consistent form (now the default) is the framework prediction covariance `PredTotal` (systematics + MC statistics, excluding the data Poisson) scaled to the custom prediction, plus the custom's own per-bin data Poisson term, i.e. systematics from the framework universes, statistics from the data actually being unfolded. This removes the 26% bias and brings all five observables to $\leq 2\%$ of the framework.

B.2 EXT (cosmic) subtraction for fake data

Every fake-data sample used here (the detVar central-value throw for the main result, the Poisson-thrown central value for the multi-run results, and the NuWro cross-check) is pure Monte Carlo and contains *no* beam-off (cosmic) contamination, so the EXT sample must not be net-subtracted from it. The framework handles this by *adding* the EXT histogram to the fake-data spectrum, so the EXT term cancels in $\text{data}_{\text{signal}} = (\text{data} + \text{EXT}) - (\text{MC bkg} + \text{EXT} + \text{dirt})$. The custom macros originally subtracted EXT (≈ 41 events at 3.283×10^{20} POT) from a sample that never contained it, a flat $\sim 3.7\%$ deficit on every observable. Replicating the framework's EXT cancellation (`fake_data_ext_cancel` in `ccpi_xsec.C`) removes it. For *real* beam-on data this term must be re-enabled, since real data does contain cosmics.

B.3 Opening-angle binning (review item C4)

The custom $\theta_{\mu\pi}$ binning used 10 reco/true bins; the framework uses 9 (the last two custom bins merged into $[2.513, 3.142]$). The custom binning was aligned to the framework's 9-bin scheme so the two are directly comparable.

B.4 Generator normalisation (NuWro vs GENIE)

In the closure test the NuWro fake data carries roughly twice the selected CC$\pi^{\pm}$ signal of the GENIE central-value prediction used to build the response, consistent with the well-documented generator spread in CC$\pi^{\pm}$ production on argon [30]. The unfolding correctly recovers the injected NuWro truth (Section C), confirming that the $\sim 2\times$ generator difference is propagated faithfully rather than absorbed into an efficiency or normalisation artefact.

B.5 Software trigger (verified to be a no-op)

A hypothesis that the absence of a `swtrig == 1` filter on MC events was causing a $\sim 13\%$ over-normalisation was investigated. The cut was implemented and the full pipeline rerun; the result was byte-identical. All MC events passing the topological cut already carry `swtrig == 1`: the $\sim 13\%$ of raw MC events failing `swtrig` also fail the topological cut. The cut has been retained for formal correctness but contributes nothing to the current normalisation.

Detector variations. The detector systematic in Table 30 of the analysis note (9.6–14.6%) was built from the Run-1 FHC detector variations applied globally. The multi-run FHC, RHC, and combined results instead use *native* Run-4 detector variations per horn mode (the standard eight-knob set), with the combined result accumulating the Run-4 FHC *and* Run-4 RHC detVar samples, each scaled to its own mode's data exposure, which the framework now supports via multi-file detVar summing. (The Run-5 FHC Recomb2 and SCE detVar productions turned out to be intrinsic-ν_e overlays with zero ν_μ content; the Run-4 FHC knobs stand in for those two.) Likewise, adding the GENIE spline weight to the central-value weight was a verified no-op: `weightSpline` $\equiv 1$ in the NuMI ntuples and the NuWro overlay's CV weights are sentinel (-1) values clamped to unity.

C NuWro Fake-Data Closure and Cross-Pipeline Validation

As a generator-model closure and an independent cross-check of the extraction, the XSecAnalyzer framework used throughout this note was additionally run (`ProcessNTuples` $\rightarrow$ `UniverseMaker` $\rightarrow$ `Unfolder`) with a NuWro overlay as *fake data* in place of the GENIE detVar-CV sample of the main results. The NuWro reconstructed spectrum is unfolded with the Genie detector response, efficiency, and Genie-predicted background subtraction, then compared to the Genie MicroBooNE-tune prediction. If the unfolding is unbiased, the ratio reflects the genuine NuWro-versus-Genie model difference rather than a procedural artefact.

Extending the framework beyond p_μ required six additional output branches in the `CC1mu1piXp` selection (reconstructed and true pion momentum, muon $\cos\theta$, and pion $\cos\theta$); the reconstructed pion momentum uses the proton-hypothesis range kinetic energy (`trk_energy_proton_v`), matching the custom pipeline. The true $\theta_{\mu\pi}$ opening angle, previously written as all-zeros, was also corrected, and all fourteen samples were re-processed.

Table 22: Configuration of the NuWro fake-data closure.

Item	Value
Fake data	NuMI FHC NuWro overlay, 6.6504×10^{20} POT native, high statistics (`onBNB`)
Exposure	all MC/EXT/dirt/detVar scaled to NuWro POT (high-statistics closure)
Response/eff.	Genie ν_μ MC overlay (central value)
Unfolding	Wiener-SVD, second-derivative regularisation
Systematics	full multisim + detector-variation universes
Observables	all five (p_μ, p_π, $\cos\theta_\mu$, $\cos\theta_\pi$, $\theta_{\mu\pi}$)

The NuWro fake data unfolds to approximately twice the GENIE MicroBooNE tune in every observable, uniformly in shape (Fig. 6, Table 23); the two independent unfolding chains recover the same ratio, cross-validating both. The flat $\sim 2\times$ offset here is a genuine NuWro-versus-GENIE-tune CC$\pi^{\pm}$ model difference in this NuMI FHC phase space: NuWro is used at its raw cross section (its `weightTune` and `ppfx_cv` branches are unset and default to unity, so no GENIE tune is applied to the fake data). This must *not* be conflated with the apparent $\sim 2\times$ seen in the primary GENIE-detVar-CV result: the two coincide in magnitude but not in cause; the primary-result excess is a POT/flux *bookkeeping* offset from using the detVar-CV sample as fake data ($1.29\times$ more signal events/POT, times the A_C smearing; Sec. B), whereas the NuWro-closure excess is a real generator difference. A caveat on the latter is that part of the excess may be NuWro's larger *background* being attributed to signal, since the procedure subtracts GENIE-predicted rather than NuWro backgrounds.

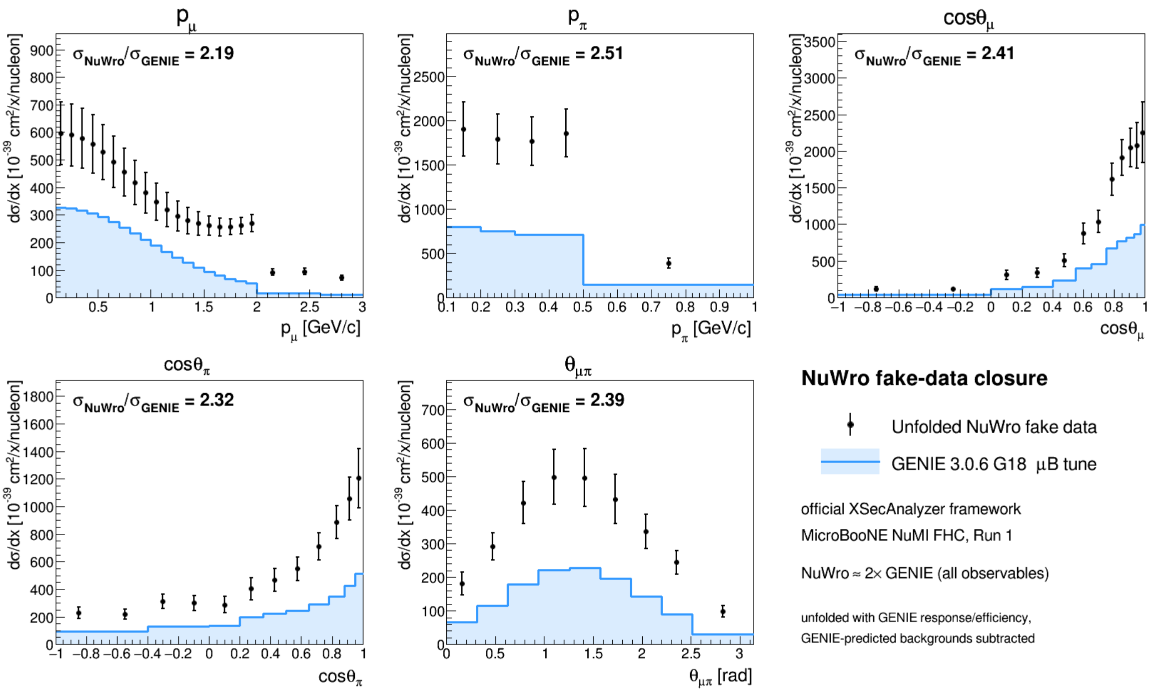


Figure 6: NuWro fake-data closure in the official XSECANALYZER framework. Points: unfolded NuWro fake data; shaded line: GENIE 3.0.6 G18 μB tune. The unfolded NuWro result sits a uniform $\approx 2\times$ above the GENIE prediction in all five observables. **This figure predates the fake-data weighting fix of Sec. 16 of the analysis note**: it was produced 2026-07-25, from a separate cross-pipeline study, and has no producer in the repository. The $\approx 2\times$ NuWro-over-GENIE ratio it shows is a generator comparison and survives the fix; the absolute normalisation of the points does not. Retained as a qualitative cross-pipeline check only.

Table 23: NuWro fake-data closure: ratio of integrated cross sections and the χ^2 of the unfolded NuWro result relative to the GENIE tune. The p_μ χ^2 is undefined because its 22-bin Wiener-SVD covariance is numerically singular (condition number $\sim 4 \times 10^{13}$).

Observable	$\sigma_{\text{NuWro}}/\sigma_{\text{Genie}}$	χ^2/bins
$d\sigma/dp_\mu$	1.137	n/a
$d\sigma/dp_\pi$	1.125	20.3/5
$d\sigma/d\cos\theta_\mu$	1.159	35.0/12
$d\sigma/d\cos\theta_\pi$	0.931	34.0/12
$d\sigma/d\theta_{\mu\pi}$	1.262	25.8/9

Technical notes. Two framework-level issues were resolved. First, the selection's opening-angle cut ($\theta_{\mu\pi} < 2.6$ rad) leaves the reconstructed bin $[2.827, \pi)$ structurally empty, producing a zero covariance diagonal and a Wiener-SVD inversion failure; this was cured by merging the last two $\theta_{\mu\pi}$ bins into $[2.513, \pi)$, giving nine bins. Second, the χ^2 computation in the diagnostic plotter was made robust to the singular p_μ covariance (it now falls back to a sentinel rather than aborting), so the closure plots render for all observables.

C.1 Extraction-bias investigation and algorithm equivalence

The fake-data closures show a consistent $\sim$1.3$\times$ ratio of the unfolded total to the raw generator truth, and the detector-variation central-value fake data sits $\sim$1.29$\times$ above the GENIE tune. Because a persistent multiplicative offset in a closure test can indicate an unfolding bug, this was investigated end-to-end; the conclusion is that **there is no framework Wiener-SVD bias**.

Algorithm equivalence. The framework unfolder (`WienerSVDUnfolder.cxx`) was compared line-by-line with the independent `RunWienerSVD_FW` implementation used for the custom pipeline. The two are algorithmically identical: covariance whitening $Q = \text{chol}(V^{-1})$, response pre-scaling $R = QA$, the SVD of RC^{-1}, the per-mode Wiener filter $W_t = \eta_t/(\eta_t + 1)$ with $\eta_t = (D_C\, V_C^\intercal C x_{\text{prior}})_t^2$, and the additional smearing matrix $A_C = C^{-1} V_C W_C V_C^\intercal C$. Feeding the framework's *exact* extracted matrices (smearceptance, prior, data, and full covariance) into a standalone replica reproduces the framework's unfolded result bin-for-bin ($\sum \hat{x} = 5836.5$ for $\cos\theta_\mu$), and the framework smearceptance column sums equal the raw efficiency.

Self-closure. The decisive test rebuilds the universes with the GENIE ν_μ overlay itself in the `onBNB` slot, so data and response come from the same sample and the detector-variation POT offset is absent by construction. The unfolded total then recovers the truth for all five observables and all three regularisations (Table 24): $\hat{x}/x_{\text{true}} = 0.94$–$1.06$. The closure holds even where the A_C diagonal is as low as 0.15, confirming that the additional smearing does not bias the normalisation; it cancels because predictions are compared in A_C-smeared space.

Table 24: GENIE-overlay self-closure: $\hat{x}/x_{\text{true}}$ for the integrated cross section, by observable and regularisation. Data and response are the same sample, removing the fake-data POT offset.

Observable	Wiener-SVD (id.)	Wiener-SVD (2nd-deriv)	D'Agostini (4 it.)
$d\sigma/dp_\mu$	1.137	0.937	1.040
$d\sigma/d\cos\theta_\mu$	1.159	0.973	0.989
$d\sigma/d\cos\theta_\pi$	0.931	0.996	0.992
$d\sigma/dp_\pi$	1.125	1.063	1.059
$d\sigma/d\theta_{\mu\pi}$	1.262	0.996	0.991

Decomposition. The $\sim 1.29\times$ detector-variation offset is a pure POT/flux bookkeeping factor from using the Run-3b detVar central-value sample as fake data (measured flat across all event types, so not a physics or generator difference); the A_C smearing cancels in the closure, and the $\sim 2\times$ NuWro/GENIE ratio (Table 23) is the genuine model difference. None of the three is an unfolding artefact. The closure χ^2 (small, high p-value) is not the most stringent test given the systematics-dominated covariance; the central $\hat{x}/x_{\text{true}}$ ratio is the more discriminating metric and is what establishes the result above. Since the extraction machinery (`ProcessNTuples`→`UniverseMaker`→`Unfolder`) is identical across horn modes, this closure validates the unbiasedness of the FHC, RHC, and combined full-exposure extractions equally.

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